PMP Key Knowledge Points

PMBOK-aligned terminology - print-friendly review edition

001 Project Definition and Characteristics

A project is a temporary endeavor to create unique products, services or results. Temporary: The project has a clear starting and end point. Temporary nature does not necessarily mean a short duration. Unique: Each project creates unique products, services or results. Progressive elaboration: continuous improvement and refinement of plans as information becomes more detailed and accurate. It enables the project management team to manage more in-depth as the project progresses.

002 Portfolio vs. Program

Portfolio: Projects or programs where project portfolios can achieve common organizational objectives. Project portfolio management focuses on prioritizing the allocation of resources through the review of projects and clusters, and on ensuring coherence between project portfolio management and organizational strategies. Project portfolio keyword: maximize value, prioritize. Program: A group of linked projects managed together to achieve results of 1+1 > 2. Program keyword: dependency.

003 Project Management vs. Operations Management

Projects and operations both involve people, limited resources, planning, execution, and control in support of organizational objectives or strategic plans. The essential difference is that projects are temporary and unique, ending when their objectives are achieved, while operations are ongoing and repetitive, sustaining normal business performance.

004 Life cycle

Life cycle judgement points: clarity of scope, frequency of delivery (one or more times). Projection type: Clear scope and low frequency of delivery. Iteration: High degree of change and low frequency of delivery. For example, the dissertation is repeated, but eventually delivered only once. Incremental type: The scope is broadly clear, but may be due to other constraints, such as resource constraints. Go to batch delivery. It's incremental. Adaptive (agile): High degree of change and high frequency of delivery.

005 Performance data

Performance data are raw observations and measurements collected from each of the ongoing activities during the implementation of the project. For example, the percentage of work done, the result of quality measurement, the date of commencement and termination of the activity, the number of requests for change, the number of deficiencies, the actual cost and the actual duration, etc. Project data are usually recorded in project management information systems and project documents

006 Work Performance Information

In. Performance data collected during the control process and analysed against project management plan components, project documents and other work performance information. For example, the status of deliverables, the status of implementation of change requests, and the projected completion have yet to be estimated. Work work performance data are collected in the course of the performance process, which is then further analysed by the control process, before generating work performance information.

007 Work Performance Reports

Performance reporting is a physical or electronic project document generated to inform decision-making, action or concern, and to compile work performance information. Performance information can be consolidated, recorded and disseminated in physical or electronic form. Performance reports are prepared in physical or electronic form based on work performance information. Examples of performance reporting include status reports and progress reports.

008 Work Performance Data / Information / Reports

Performance data are readily recorded at the time of project implementation and are first-hand information. Work performance information is obtained during various monitoring processes and is based on actual and planned deviations, which are analysed in order to predict the future. Performance reporting is a project-wide level, with a comprehensive comparison of implementation and plans to determine whether action is required and generally applicable to client management.

009 Enterprise environmental factors vs. organizational process assets

Environmental factors are conditions beyond the control of the project team that will affect, restrict or direct the project. Objective presence; potential help or hindrance; project manager has to comply with cumulative (updating and adding) organizational process assets; useful for the future; project manager has an option to use (tailorable)

010 Comparison of organizational structure

Functional type: There is generally no project manager and the role of coordinator. Weak matrix: There is a project manager, but the project manager has less authority than the functional manager. Strong matrix: Project managers have higher rights themselves. Balance matrix: The rights of the two are equal and when resources are needed, consultations are held with functional managers. Project type: The largest project manager.

011 PMI Talent Triangle

Technical project management: knowledge, skills and behaviour related to specific areas of project, portfolio and portfolio management, i.e. technical aspects of role fulfilment. Leadership: the knowledge, skills and behaviours required to guide, motivate and lead a team can help the organization achieve its operational objectives. Strategic and business management: Knowledge and professional skills on industry and organization can help improve performance and achieve better business results.

012 Project Management Office (PMO) (PMO) (PMO) (PMO) (PMO) (PMO) (PMO)

PMO is an organizational structure that standardizes project-related governance processes and promotes the sharing of resources, methodologies, tools and technologies. The terms of reference of PMO could be small, ranging from the provision of project management support services to the direct management of one or more projects. PMO has several different types: support, control, command

013 Five process groups for project management

Start: Define a new project or a new phase of an existing project and authorize the start of the project or phase. Planning: Clear project scope, definition and optimization of objectives. Implementation: Complete the work identified in the project management plan to meet project requirements. Monitoring: Tracking, reviewing and adjusting project progress and performance, identifying necessary changes in the plan and initiating corresponding changes. Closure: the process undertaken to formally complete or close the project, phase or contract.

014 Project Phase vs. Process Group

Project phases divide the project by schedule position and life cycle stage so the team can manage and control progress. Process Groups are not project phases. Each phase or subproject may repeat the relevant Process Groups. A phase describes where the project is in its life cycle; a Process Group describes the management processes being performed.

015 Business case

Business cases are used to validate the value of projects and typically include business needs and cost-benefit analyses. (b) The business case or similar document provides the necessary information from a commercial point of view to determine whether the project is worth investing;

016 Project benefits management plan

The project benefit management plan describes the manner and time in which the benefits will be realized by the project, as well as the benefit measurement mechanisms to be developed. Project benefits are the result of actions, behaviours, products, services or outcomes that create value for the initiating organization and the intended beneficiaries of the project. Target benefits should be identified early in the project life cycle and benefit management plans developed accordingly. The development of benefit management plans requires the use of data and information from business demonstrations and needs assessments.

017 Project selection and initiation

A simple understanding is that project management is "doing the right thing first and then doing the right thing", in which "doing the right thing" means that feasibility studies, analysis of the background of the project, etc., are required before the project is launched to determine whether it is worth doing. Currently, there are a number of project options, which can be divided into two main categories, namely, benefits measurement and constraints optimization.

018 Feasibility study

The task of the feasibility study is to determine, at the lowest cost, whether the problem can be solved in the shortest possible time, not to solve the problem, but to determine whether it is worth addressing. To achieve this, it is necessary to analyse the advantages and disadvantages of several major possible solutions to determine whether the intended product objectives and sizes are realistic and whether the benefits that can be derived from the completion of the product are such that it is worth investing in.

019 Development of project charters

The development of the project charter is the process of preparing a document formally approving the project and authorizing the project manager to use organizational resources in project activities. The main role is to clarify the direct link between the project and the strategic objectives of the organization, establish the formal status of the project and demonstrate the organization ' s commitment to the project. The approval of the project charter marked the official launch of the project. Project managers should be identified and appointed as early as possible, preferably at the time the project charter is prepared, and at the latest before planning begins.

020 Main elements of the project charter

The purpose of the project or the rationale for approving the project; measurable project objectives and associated success criteria; high-level requirements; High-level project descriptions, boundary definitions and key deliverables; Major risks to the project; overall milestone progress plan; overall budget; list of key stakeholders; Project approval requirements; project exit criteria; assigned project managers and their responsibilities and authority;

021 Role of the project charter

The existence of the project is officially announced. The project manager is formally appointed and is authorized to carry out project activities using the resources of the organization. The project charter is issued by management and the project manager is the implementer of the project charter. The project charter should provide for larger, principle-based issues, and normally no changes to the project charter are required as a result of a change in the project, and if so, who issues the change?

022 Assuming that the log vs. assumes vs. constraints

Assumptions log: used to document all assumptions and constraints throughout the life cycle of the project. Assumptions: Some of the conditions that are assumed to be "real" or "determined" are an important basis for project planning, and if the conditions are wrong, everything that follows will follow. Constraints: Constraints limiting the choice of project teams, most commonly in terms of scope, time, cost, quality, human resources, etc.

023 Development of project management plan

The development of the project management plan is the process of defining, preparing and coordinating all components of the project plan and consolidating them into an integrated project management plan. The primary role is to produce a synthesis document that will be used to determine the basis of all project work and how it will be implemented. The project management plan determines how the project is implemented, monitored and closed. The development of a project management plan requires the integration of a range of related processes and their continuation to the end of the project.

024 Project management plan

The project management plan is an integrated plan and results from the integration of a series of sub-management plans and other elements. Once the project management plan has been established as a baseline, it can be changed only after a request for change has been submitted and approved for implementation of the overall change control process. Project management plans may need to be approved not only by management but also by other major project stakeholders.

025 Elements of a project management plan

Sub-management plan: scope management plan, needs management plan, progress management plan, cost management plan, quality management plan, resource management plan, communication management plan, risk management plan, procurement management plan, related party engagement plan. Baseline: Scope baseline, progress baseline, cost baseline. Other components: change management plan, configuration management plan, performance measurement baseline, project life cycle, development methodology, management review, etc.

026 Benchmarks for project management plans

[Baseline] is an approved project plan plus or minus approved changes to serve as a basis for comparison, whereby project performance is assessed as good and bad and actual performance is determined within acceptable deviations. [common benchmarks] include specific scope benchmarks, benchmarks for progress, and cost benchmarks.

027 Launch of the vs. Conference

The Initiating Meeting was the end of the start-up phase, prior to the approval of the project charter, and the project management plan had not been prepared to issue the project charter, indicating that the project could start planning. The Project Management Plan (PMP) was prepared by kickoff meeting to communicate project objectives, clarify the roles and responsibilities of each of the parties involved and build team confidence in the commitment of the project.

028 Directing and managing project work

The direction and management of project work is to lead and implement the work identified in the project management plan to achieve project objectives and to implement the approved change process. The main role of the process is to improve the likelihood of project success through integrated management of project work and deliverables. The process requires the project manager and the project team to take action to implement the project management plan to achieve the project objectives

029 Deliverables

Deliverable results are any unique and verifiable product, outcome or service capacity that must be produced when a process, phase or project is completed. It is usually the result of a project and may include components of a project management plan. Deliverable results are measurable, verifiable, tangible or intangible project results that must be submitted in order to complete the project.

030 Fulfillable results

- Transfer of the final product, service or outcome (in the end of the phase, intermediate product, service or result) of the project's output through the closure of the project or phase.

031 Change Node

With the endpoint, once the project is accepted and accepted, the new requirements presented need to be changed to be completed later. Changes can be proposed at the end, but most cases will not be approved. When at the end of the period the client proposes minor changes, it is best to persuade that there is no need or to study alternatives. Clients proposed major changes and proposed new contracts. The changes proposed by the client were made by the development team.

032 Requests for change

The change request is a formal proposal to modify any document, deliverables or benchmarks. Any party involved in the project may submit a request for change, which should be reviewed and processed through the implementation of the overall change control process. A request for a change may come from within or outside the project; it may be optional or mandatory by law or agreement. Requests for change may include: preventive measures, corrective measures, remedy for defects, updates

033 Preventive measure vs. correction of vs. deficiency Help!

Preventive measures: purposeful activities to ensure that future performance of project work is in line with the project management plan. For possible future deviations. Corrective measures: purposeful activities to realign project performance with project management plans. In response to actual deviations. Deficiencies: purposeful activities to amend inconsistent products or product components. Targeting only the quality of projects.

034 Tools and techniques — expert judgement

Expert judgement may come from any group or individual with expertise or professional training experience and may be obtained through a number of channels, including other departments within the organization, consultants, stakeholders (including clients or originators), professional and technical associations, industry associations, subject matter experts, PMO. Keyword: "Professional, expert group, complex and historical data not available".

035 Assumptions Log

The assumptions logs are used to document all assumptions and constraints throughout the life cycle of the project. Typically, high-level strategic and operational assumptions and constraints are identified when preparing business cases prior to project start-up. These assumptions and constraints should be incorporated into the project charter. Lower-level activities and mission assumptions are generated during the project as activities such as defining technical specifications, estimates, progress and risks are carried out.

036 Change Log

The change log contains the status of all requests for change throughout the project or phase. The change log is used to communicate changes to affected parties, as well as approval, deferral and rejection of requests for changes. The change is usually related to the specific party concerned, as the party concerned may be the author of the request for the change, the author of the request for the change or the person affected by its implementation.

037 Problem Log

The issue log is a project document that records and follows up on all issues, and the required documentation and follow-up may include: the type of problem; the question person and time; the description of the problem; the priority of the issue; who is responsible for resolving the problem; the target date for resolution; the status of the problem; and the situation of final resolution. Issue logs can help project managers to effectively follow up and manage issues and ensure that they are investigated and resolved.

038 Register of lessons learned

The register of lessons learned could contain the type and description of the situation, and the register of lessons learned could also include situation-related implications, recommendations and courses of action. A lessons-learned register can record challenges, problems, perceived risks and opportunities, or other applicable elements. At the end of the project or phase, relevant information is incorporated into the knowledge base of lessons learned as part of the assets of the organizational process.

039 Tools and techniques - Brainstorm

Brainstorms, designed to generate a large amount of creativity within a short period of time, apply to the team environment and require guidance from the lead. Brainstorms consist of two components: creative creation and creative analysis. Project charters can be developed through headstorms to collect data, solutions or ideas from stakeholders, subject matter experts and team members. Keywords: "Creativity, innovation, discussion on gathering new ideas".

040 Tools and technologies — guidance

Guidance is the ability to effectively guide team activities to success in reaching decisions, solutions or conclusions. The guide ensures the effective participation of participants, mutual understanding, consideration of all views, full support for the conclusions or results obtained in accordance with the established decision-making process, and the subsequent rational implementation of the action plans and agreements reached. Keywords: "A disagreement, in order to reach agreement" etc.

PMP Key Knowledge Points

PMBOK-aligned terminology - print-friendly review edition

041 Visible knowledge vs. Invisible knowledge

Visible knowledge: anyone who can be expressed in words and numbers and in information, scientific law, specific specifications and manuals. This knowledge can be transmitted among individuals at all times. Invisible knowledge: something that is fairly personal and flexible, which varies from person to person, is difficult to describe in formulas or words and therefore difficult to pass or share. It's like an individual's subjective insight, intuition and premonition.

042 Concept of monitoring project work

Monitoring of project work is the process of tracking, reviewing and adjusting project progress to meet the performance targets set in the project management plan. Oversight is one of the project management activities throughout the project cycle, which includes the collection, measurement and dissemination of work performance information, the analysis of measurements and the forecasting of trends in order to drive process improvement. Controls include the development of corrective or preventive measures or re-programming and follow-up on the implementation of action plans to ensure that they are effective in addressing problems.

043 Differentiation of several analytical techniques

Analysis of options: A combination of corrective or corrective and preventive measures to be implemented in case of deviations. Cost-benefit analysis: helps identify the most cost-effective corrective measures in case of project deviations. Analysis of deviations: review of differences (or deviations) between target and actual performance, which may relate to estimates of duration, cost estimates, use of resources, resource rates, technical performance and other measurements

044 Trends analysis vs. earned value analysis

Trend analysis: Forecasting future performance on the basis of past results, it predicts delays in the progress of its projects and alerts project managers in advance of problems that may arise at a later stage of progress in accordance with established trends. A trend analysis should be conducted sufficiently early for the project to allow the project team time to analyse and correct any anomalies. The results of the trend analysis may be used to recommend the necessary preventive measures. Profit analysis: The earned analysis provides a comprehensive analysis of scope, progress and cost performance.

045 Implementation of overall change control

The overall change control process is implemented throughout the project. Any interested party in the project may request a change. All requests for change must be recorded in writing. The analysis and review of requests for change must be comprehensive and comprehensive and must examine the possible impact of each change on all aspects of the project. Review all requests for changes, all recommended corrective and preventive measures and approve or reject them. Ensure that only approved changes are implemented.

046 Change Control Committee

The Change Control Committee (CCB) is a body responsible for approving and approving project changes. The main function of the CEB is to provide guidance for the preparation of requests for change, to evaluate requests for change and to manage the implementation of approved changes. The organization could include the main project stakeholders in the committee and, depending on the particular needs of each project, could also include several project team members

047 Basic principles of change control

The basic principles of project change control are summarized as follows: 2. Strict control over the submission of project change requests; 3. Give high priority to scope changes; 4. Signature of agreement to change control; 5. Implementation of changes based on benchmarks; 6. Need to be supported by good change control tools; 7. Integration of project changes into project management plans; 8. Timely issuance of change information.

048 Authority to approve changes

Changes to the project charter are authorized only by those (usually management) who sign or approve it. If the change is in the project management plan (which does not change the project baseline), the project manager has the authority to make decisions. Only the CCS would be authorized if the change would affect the scope, time, cost and quality objectives of the project, which would result in a change in the project baseline. Changes in emergency situations may be effected without approval, pending subsequent follow-up.

049 Change process

In general, changes proposed for external parties: Request for change - >Receipt of change - >Identification of change (know what change) - >Evaluating the impact of change on the project - >Seek options for processing change - >Requirement of comments from project stakeholders - >Approval or rejection of change - >Enforcement of change (if approved) - >Track of change implementation - >Information communication archive

050 Treatment of changes in different states

Treatment of changes in different states: 1. What happens once the changes are approved? - - Update the project management plan, update the change log and then implement the specific changes. - Update the change log. What do you do with the change without going through the change process?

051 Ending Event Order

The main sequence of closing events [tailorable on the basis of the project and not entirely based on this): 1, handover of results; 2, financial closure; 3, satisfaction surveys; 4, register of lessons learned; 5, document update (organizational process asset update); 6, final report; 7, document filing; 8, celebration; 9, team dissolution.

052 End of contract vs. project

The contract was concluded by agreement between the project team and the buyer to check whether all the requirements of the contract had been fulfilled and whether the project had been terminated, i.e. the project acceptance. The administrative closure, also referred to as the management closure, is internal in nature and will be archived, including through the filing of completed project files, the public declaration that the project has been closed and the related product description has been transferred to the protection team, along with lessons learned.

053 Tool technology - multi-standard decision analysis

The multi-standard decision analysis technology builds on the decision matrix and uses a systematic analysis methodology to establish a variety of criteria, such as risk levels, uncertainty and value gains, to assess and sequence a wide range of programmes. The decision matrix, also known as the decision table, the profit and loss matrix, the profit and loss statement and the risk matrix, is a matrix indicating the interrelationship between the decision-making programme and the relevant factors and is often used for quantitative decision-making analysis.

054 Multi-standard decision analysis - priority matrix

It can be used to identify key issues and appropriate options and prioritize options through a range of decisions. First, the criteria are sequenced and weighted, then applied to all options, calculating the mathematical score for each option, and then ranking the options according to the score. Multi-standard decision analysis techniques were used to implement overall change control, collect demand, define scope, plan quality management, manage quality, access resources, plan risk response, etc.

055 Project management information system

The Project Management Information System (PMIS) provides information technology (IT) software tools such as the progress plan software tool, the work authorization system, the configuration management system, the information collection and dissemination system, and interfaces with other online automated systems, such as the corporate knowledge base. Automatic collection and reporting of key performance indicators (KPI) can be a function of the system.

056 Project scope vs. product range

Product range refers to the characteristics and functions of the product, service or outcome. The scope of the project refers to work that must be performed to deliver a product, service or result that has the required characteristics and functions. The product range is the basis of the project scope, which is defined as the measure of the product requirements, while the project scope is defined as the basis for the generation of the project management plan, and the two ranges are different in application

057 Definition Scope vs. Recognition Scope

The definition of scope is the process of developing detailed descriptions of projects and products. The main role is to describe the boundary and acceptance standards for products, services or outcomes. The scope of the definition is after collecting the demand, simply defining what the project is offering. The recognition of scope is the formal acceptance and acceptance of deliverables from completed projects. The primary role is to make the acceptance and inspection process objective; at the same time, by identifying each deliverables. In short, verification of deliverables.

058 Scope management plan

The scope management plan describes how the project scope will be defined, developed, monitored, controlled and validated. (c) WBS based on the detailed project scope statement;

059 Needs management plan

The needs management plan describes how the needs of projects and products will be analysed, documented and managed. How to plan, track and report on needs activities; Configure management activities; Prioritization of needs; Measuring indicators and the rationale for their use; Reflecting which demand attributes will be included in the tracking structure of the tracking matrix.

060 Definition and classification of collection needs

Collecting demand refers to defining and managing client expectations. Demand is the foundation of the WBS, and cost, progress and quality plans are based on these needs. Demand is divided into business needs, related party needs, solution needs (functional/non-functional), transition and readiness needs, project needs and quality needs.

061 Technologies for collecting demand tools - prototype approach

The prototype approach is a way to quickly build a product model for presentation to stakeholders, based on the initial needs of the parties concerned and using product development tools, and to communicate with the parties concerned on the basis of this approach, leading to the rapid development of the products required by the parties. The prototype approach is consistent with the concept of progressive elaboration and, after sufficient feedback loops, allows for a sufficiently complete demand from the prototype to proceed to the design or manufacturing stage.

062 Demand tracking matrix

The Demand Tracking Matrix is a table linking product demand from its sources to deliverables that meet demand. The use of a needs-tracking matrix, which links each demand to operational or project objectives, helps to ensure that each demand has a commercial value. These include: business needs, opportunities, goals and objectives; project objectives; deliverables in project scope/WBS; product design; product development; testing strategies and testing scripts; high-level demand to detailed needs.

063 Project scope statement

The project scope statement describes project scope, key deliverables, assumptions and constraints. It records the entire scope, including the project and product range; describes in detail the deliverables of the project; and represents the consensus reached among project stakeholders on the scope of the project. These include: product frame description; acceptance criteria; deliverables; and liability except for projects.

064 Request tracking matrix vs. project scope statement

The Demand Tracking Matrix is a table linking product demand from sources to deliverables that meet demand. Scope of application: Linking needs to deliverables or products to the plan needs to be determined if needs are met. The project scope statement describes project scope, key deliverables, assumptions and constraints. Scope of application: Where deliverables can be found, or specific, detailed, the most accurate estimates emerge and require the most detailed documentation.

065 Scope benchmarks

The scope baseline, which is an approved scope statement, WBS and the corresponding WBS dictionary, can be changed only through a formal border control procedure and is used as a basis for comparison. The scope baseline is an integral part of the project management plan. The scope baseline includes: project scope statement, WBS, work package, planning package, WBS dictionary.

066 Needs analysis

Demand analysis is the extraction, analysis and careful examination of the acquired needs to ensure that all project stakeholders understand their meaning and identify errors, omissions or other deficiencies. The key to needs analysis is research and understanding of operational areas. (b) Creation of prototypes;

067 WBS & WBS Dictionary

WBS:WBS is a hierarchy of the scope of work that the project team needs to implement in order to achieve project objectives and create the required deliverables. The break-down structure is broken down into one layer and represents a more detailed definition of project work. WBS Dictionary: A document detailing deliverables, activities and progress information for each component of the WBS.

068 Scope spread

1 The scope of the judgement spreads: the outside team requires additional functionality, the team does not go through the process, the direct execution is spread. 2. Auto-enhancement within the team is gold plating. 3 Quest description: Direct addition function - spread of presence should first analyse the impact and stop adding or moving to change process.

069 Create Job Decomposition Structure

The creation of a decomposition structure (WBS) is the process of decomposing project deliverables and project work into smaller, more manageable components. The main function of the process is to provide the structure for the content to be delivered, which is conducted only once or only at the predefined point of the project. The next step after the completion of the scope of the definition or the scope statement is to create a decomposition structure.

070 Create WBS Tools - Decomposition

Decomposition is a technique that gradually divides project scope and deliverables into smaller, more manageable components. Activities required for disaggregation: Identification and analysis of deliverables and related work; Determination of the structure and organization of WBS; Top-down and bottom-up refinement of disaggregation; Development and allocation of marking codes for WBS components; Verification of the adequacy of decomposition;

071 Create WBS Tools - Decomposition Principles

100% rule: WBS contains all product and project work; Rolling planning: deliverables or sub-projects that are to be completed only in the future may not be broken down at this time. (b) In general, WBS should be kept on the 4-6 floors;

072 Control account vs. packages

The control account is a management control point in which scope, budget (resource plan), actual costs and progress are consolidated and compared with earned value to measure performance; a control account may include one or more work packages, but each workpack can only be a control account. The work packages are the deliverables or components of project work at the bottom of each WBS branch.

073 Planning package

The planning package is under the control account and works are known but detailed progress activity components of the WBS. The planning package is a WBS element under the control account and above the work package. The planning package is used for the time being. As the situation becomes clearer, planning packages will eventually be broken down into work packages and corresponding specific activities.

074 Confirm range

The scope of confirmation is for the client or sponsor to review the verifiable deliverables exported from the quality control process and confirm that these deliverables have been successfully completed and formally accepted. The process recognizes and ultimately accepts deliverables. The main role is to make the acceptance and inspection process objective and, at the same time, to increase the likelihood of receiving and inspection of final products, services or results by identifying each deliverables. The scope should be established throughout the project.

075 Confirm range vs. control quality

The identification of scope is primarily concerned with acceptance of deliverables, quality control is primarily concerned with the validity of deliverables and whether quality requirements are met. Quality control usually takes place before the confirmation scope, but both can be performed simultaneously. Quality control is internal and is implemented by the corresponding quality department of the implementing organization; the scope of confirmation is performed by external stakeholders (clients or sponsors) on the deliverables of the project

076 End of range vs. project

The difference between the identification of scope and project closure is that while both the recognition of scope and the project closure are carried out at the end of the phase, the recognition of scope emphasizes the verification and acceptance of deliverables, while the closure of the project emphasizes the process work to be done to close the project (or phase). Both the scope and the end of the project are recognized for receiving and inspection, with an emphasis on the deliverables of the receiving and inspection project and on receiving and inspection products.

077 Validate Scope vs. Verify Product

Product verification focuses on whether the product has been completed and is usually performed by the sponsor or customer at the end of a project or phase. Validate Scope focuses on formal acceptance of project deliverables. Its primary concern is whether the customer or sponsor accepts the deliverables produced by the project.

078 Control range

The control scope is the process of monitoring the scope status of the project and products and managing the change in the scope baseline. The main function of this process is to maintain the maintenance of the scope baseline throughout the life of the project. Controlling the scope of the project requires ensuring that all requested changes, recommended corrective or preventive measures are processed through an overall change control process.

079 Tool technology -- interviews

Interviews are a formal or informal way of obtaining information through direct conversations with the parties concerned. The interviews were structured and non-structured. Structuralization refers to the preparation of a series of questions in advance and to their targeting; rather than structuralization, it is simply a rough idea, based on the specifics of the interviews. The most effective interviews are conducted (semi-structured) in combination with these two approaches, and after all, it is impossible to plan everything together and to maintain good flexibility.

080 Typical interview practices and applications

Interviews are typically conducted by posing pre-set and improvising questions to respondents and documenting their responses. Usually in one-on-one form, but also with multiple respondents and/or visitors. Interview techniques were used in eight processes, including the development of project charters, project management plans, collection of needs, planning quality management, risk identification, implementation of qualitative risk analysis, implementation of quantitative risk analysis and planning of risk response.

081 Tools Technology - Focus Group

Focus groups are intended to bring together pre-defined stakeholders and subject matter experts to learn about their expectations and attitudes regarding the products, services or outcomes under discussion. The focus group session was led by a trained moderator for interactive discussions. Focus group meetings, which are group interviews rather than one-on-one interviews, tend to be more interactive than one-on-one interviews.

082 Tools - Nominal Group Technologies

Nominal group technology is a technique used to promote brainstorms, and is the most useful innovation through voting to further brainstorms or prioritization. Nominal group technology is a form of concoction of brainstorms, which, compared to general brainstorms, allows those who are not very good to express their views. A typical small group of technology is Delphi. Only notional group skills were used in the collection of demand.

083 Delphi technology

Delphi technology is linked and distinct from the usual method of convening experts to meet, collectively discuss and arrive at unanimous forecasts. Delphi technology can serve the merits of the expert meeting approach, namely: to play the full role of experts, to pool their ideas and to be accurate;

084 The vs. disadvantages of Delphi technology

Delphi technology is linked and distinct from the usual method of convening experts to meet, collectively discuss and arrive at unanimous forecasts. (a) Delphi technology avoids the shortcomings of expert meeting methods:

085 Tools - questionnaires and surveys

Questionnaires and surveys refer to the rapid collection of information from a large number of respondents through the design of written questions. The survey methodology is best suited to the diversity of the audience, the need for rapid completion of the survey, the geographical dispersion of the respondents and the need for statistical analysis. Questionnaires and survey techniques were used in three processes: collecting demand, controlling quality and identifying stakeholders.

086 Survey vs. interviews

Compared to interviews, the questionnaire can be collected from a large number of responses in a short period of time at a low cost; the questionnaire allows respondents to fill in anonymously, and most stakeholders may provide real information; and the results of the questionnaire are better organized and statistically collected. The greatest deficiency of the questionnaire was the lack of flexibility. Better practice is to combine interviews and questionnaires. The results of the analysis are supplemented by a questionnaire followed by small-scale stakeholder interviews.

087 Tool Technology - Stick Contrast

Benchmarks refer to the comparison of actual or planned products, processes and practices with those of other comparable organizations in order to identify best practices, generate improvements and inform performance appraisals. Projects that serve as poles can come from within or outside the implementing organization, or from the same application area or other application area, and are also allowed to be compared with projects in different applications or industries. Benchmarks were used for collecting demand, planning quality management and participation of stakeholders in planning.

088 Acronym vs. mind map

Prosthetics. The technology used to group a large number of ideas for further review and analysis. It's a bunch of ideas (e.g., easy to post) that come together in a certain way, at the very end of the day. Thinking. Combining ideas from brainstorms into a map that reflects commonalities and differences between ideas and stimulates new ones.

089 Milestones

Milestones are important points or events in the project, and the list of milestones lists all project milestones and indicates whether each milestone is mandatory (e.g., contractual requirements) or selective (e.g., based on historical information). The milestones last for zero because they represent an important point or event. The list of milestones sets out the planned target dates for specific milestones to be used to check whether the planned milestones have been met.

090 Define Activity

The definition of activities is the process of identifying and documenting specific actions required to achieve the deliverables of the project, with the primary role of disaggregating work packages into activities as a basis for estimating, planning progress, implementing, monitoring and controlling the work of the project. The WBS creation process has identified the bottom deliverable of the WBS, the work package. The work package should normally be further subdivided into a smaller component, i.e. activities, representing the work input required to complete the work package.

091 Split vs. rolling planning

Decomposition refers to the subdivision of final results into smaller, more manageable modules to better manage and control, where final results refer to activities rather than deliverables. Rolling planning is an iterative planning technique, that is, detailed planning of work to be done in the near future, with rough planning of forward work at a higher level. It is a gradual and detailed planning formula that applies to work packages, planning packages and distribution planning using agile or waterfall approaches.

092 Progressive, detailed vs. rolling planning

Gradual detail is that it may not be clear at first, and further analysis will need to be done through multiple rounds, and then the iterative process will slowly become clear. For example, phase I, phase II and phase III, each of them is more detailed than the previous phase, which is a gradual detail. Rolling planning is an incremental and detailed planning approach and an iterative planning technique. Gradual detail is a process, and a rolling plan is a gradual and detailed planning approach.

093 Estimated duration considerations

Factors to be taken into account in estimating duration include: 1, quantity of resources; 2, skill proficiency; 3, technological advances; 4, employee incentives; 5, pattern of diminishing returns: The addition of a factor (e.g., resources) to determine the input required for unit outputs would eventually reach a critical point where the output or output after that point would decrease as the factor was added.

094 Mandatory vs. selective dependence

Mandatory reliance, also known as hard logic or hard reliance, is a relationship of dependence required by law or agreement, or determined by the intrinsic nature of the work, which exists between the activities themselves and cannot normally be changed by the project team. Selective dependence, also known as soft logic, preferential logic, and freely manageable dependency, is a logical relationship based on the best practice of the sequence of activities in a particular area of application or project, which is artificially prioritized.

095 External dependency vs. internal dependency

External dependency relates to the relationship between the project and non-project activities and often depends on the logic of any third party outside the project, such as approval by the government department, supply of equipment suppliers, etc. Internal dependency is a close forward relationship between project activities, usually under the control of the project team. For example, only when the machine is assembled can the team test it. It's an internal.

096 Advance vs. lag

Advance amounts are those that are based on the time of completion or commencement of the immediate activity, which can be initiated or completed earlier. Use of time in advance to allow for early start of late activity. The lag is the amount of time that must be postponed for the commencement or completion of the immediate activity, which is based on the completion or commencement of the activity. Using time lags, it is possible to delay the start of late activity.

097 Progress network analysis

The progress network analysis is an integrated technique for the creation of project progress models, using several other techniques. For example, the key path approach, the key chain approach, resource optimization techniques, modelling techniques (scenario analysis and resource balance) are the earliest and latest starting dates for the uncompleted components of the project activity and the earliest and latest completion dates.

098 Critical Path Method vs. Key Chain Method

In the critical path approach, only the relationship of dependence between activity and activity is considered, rather than the relationship of activity to resources. The key chain approach allows the project team to set buffers on any project progress path to address resource constraints and project uncertainty. The key chain approach, on the other hand, identifies the key path, then adjusts the network map to the resource constraints and identifies the key chain. In other words, the key path to resource constraint is the key chain.

099 Critical Path Method - Floating Time

The total floating time is the mobile time of the activity without delay. The total floating time of the activity is equal to the difference between the time the activity is completed at the latest and the earliest, or between the beginning of the activity and the earliest. Project floating time means the time a project can delay without affecting the completion date required by the outside world (e.g. client or management). The total floating time on the key path is 0.

100 Total floating time vs. free floating time

Total Float Time (Total Float): The total amount of time that a progress activity can delay without delaying the project completion date or violating progress constraints (the amount that can be postponed at the earliest). Free Float: the amount of time that a progress activity can be postponed without delaying the earliest commencement date of its immediate progress activity (the earliest completion time can be postponed)

PMP Key Knowledge Points

PMBOK-aligned terminology - print-friendly review edition

101 Forward Relationship Drawing Method

The Advanced Relationship Mapping (PDM) is a technique for creating a progress model, with nodes for activities and one or more logical connections to show the sequence of activities. Straight ahead activities: activities that rank ahead of an activity in the logical path of the progress plan. Activities in the immediate future: Activities in the logical path of the progress plan that are placed behind an activity. This includes four types of dependency or logic: from completion to completion, from beginning to beginning, from beginning to completion.

102 By analogy estimates

Practice: Estimating similar parameters or indicators for future projects on the basis of past parameters values (e.g. duration, budget, size, weight and complexity). It's a rough estimation method. Scope of application: This technique is often used to estimate the duration of the project when the details of the project are insufficient. Comparable estimates use a combination of historical information and expert judgement. Characteristics: Comparive estimates are usually lower cost and less time-consuming, but also less accurate.

103 Parameter estimation

Practice: Use statistical relationships between historical data and other variables to estimate activity parameters such as cost, budget and duration. Characteristics: The accuracy of parameter estimates depends on the maturity of the parameter model and the reliability of the underlying data. Common methods: Common parameter estimation methods are regression analysis and learning curves.

104 Three points

Impact: By taking into account uncertainties and risks in the estimation, the accuracy of the estimate of the duration of the activity can be improved, indicating the range of changes in the estimate of duration. An approximation zone that facilitates the definition of duration. Formula: Most likely time (tM), most optimistic time (tO), most pessimistic time (tP), most desired duration (tE) $tE = (tO + tM + tP) / 3$ Beta distribution (from Program Evaluation and Review Technique (PERT)) $tE = (tO + 4tM + tP) / 6$ (default)

105 Resource balance vs. resource smoothing

Resource optimization techniques include resource balance and resource smoothing. Resource balance: Use scenario: resources are available at a given time, resource quantity limits, resources are overallocated. Time used: Resource smooth before critical path is determined. Targeting: Targeted generally for critical resources. Resource smoothing: Use scenario: uneven resource utilization, exceeding the level of resources targeted. Time used: normally carried out after balance of resources. Target audience: Generally for non-critical resources.

106 We're on vs. quick

Resource compression techniques are those that shorten or accelerate progress periods to meet progress constraints, mandatory dates or other progress targets without reducing the scope of the project. This includes: rush work and quick follow-up. (c) A technique to compress the schedule by increasing resources at minimal cost. Examples: approval of overtime, additional resources, etc.

107 Project Calendar vs. Resource Calendar

Project calendar: The project calendar refers to the calendar of working days and work shifts available to show progress activities, distinguishing the time periods (in days or smaller units) available for progress activities from those that are not. Resource calendar: The resource calendar is the calendar of available working days or work shifts for each specific resource. The resource calendar identifies working days, shifts, normal business hours, weekends and public holidays for each specific resource availability.

108 Analysis of reserves

The reserve analysis, also known as set-aside management, is used in progress, cost and risk management to identify the basic features of the components of the project management plan and their interrelationships, thereby creating a reserve for the project ' s duration, budget, cost estimates or funding requirements. The reserve analysis is used to determine the contingency and management reserves required for the project. Reserve analysis techniques were used in five processes, including estimating the duration of activities, estimating costs, budgeting, controlling costs and monitoring risks.

109 Scenario analysis

Scenario analysis is an assessment of scenarios and predicts their impact on project objectives (positive or negative). Scenario analysis is an analysis of "what would the situation be if scenario X appeared?" is an analysis of the question of "how should a given scenario be addressed when it occurs". Scenario analysis techniques have been used to plan and control progress.

110 Progress benchmarks

The progress benchmark is an approved progress model that can be changed only through a formal change control procedure and is used as a basis for comparison of actual results. It is accepted and approved by the parties concerned and includes the start and end of the baseline. In the process of controlling progress, the actual start and end time will be compared with the approval time to determine whether there is a deviation

PMBOK-aligned terminology - print-friendly review edition

111 Progress data

Progress data are a collection of information used to describe and control progress plans, including, at a minimum, progress milestones, progress activities and activity attributes, as well as all known assumptions and constraints, and are often used as supporting information, including: resource requirements designed in time, alternative progress plans, contingency reserves for progress.

112 Project progress plan

The project progress plan is the output of the progress model, demonstrating the interrelationship between activities, as well as the planning time, duration, milestones and resource requirements. The project schedule must include at least the planned start and end of each activity. The project progress plan is commonly used as a network chart, a cross-road chart and a milestone map, indicating that these three forms of progress plan constitute a hierarchy from the lower (detailed) to the upper (broad) levels.

113 Transverse Chart

The trajectories are also referred to as Gantt or Bar Charts, and the progress activity is shown on the vertical axis, with the date lined in the cross-axis, and the duration of the activity indicates a horizontal bar (traverse) positioned by the start and end dates. It combines the two functions of planning and scheduling. The working phase of the activity is expressed horizontally, the beginning and end of the segment corresponds to the beginning and completion of the activity, respectively, and the length of the segment indicates the time required to complete the activity.

114 Milestone

The milestone map is similar to the trajectories, but it shows only the main deliverables, as well as the start and completion dates of key external interfaces, without the need for a breakdown of activities. In practice, milestone maps are commonly referred to as Master Schedule, control schedule or level-I schedule. The milestone is the mark of the completion of a series of activities, the end of a certain phase, and it has no schedule.

115 Project progress network chart

Adding graphics of the activity date information generally indicates both the project network logic and planned activities on the project ' s critical path. The progress map is presented in a variety of forms, for example, single-name networks, logical cross-road maps, timescale networks, etc. Could be a simple pure logical map. It is also possible to add information on the estimated duration to the calendar after the web chart is drawn to show the planned start date of each activity and the planned completion date.

116 Control progress

Content: To determine the current state of progress of the project; To influence the factors that give rise to the change in progress; To reconsider the necessary schedule; To judge whether there has been a change in the progress of the project; To manage the change when it actually occurs; Practice: If there is a problem with describing the progress, the need to control it first; To judge the impact of the progress, whether it needs to change or, if so, to move through the change control process;

117 Tools Technology - Performance Review

Performance reviews are a technique for measuring, comparing and analysing the actual performance of the work being carried out against benchmarks. Performance reviews are used in four processes: control of progress, control of quality, control of resources and control of procurement. Controlling progress, for example, means measuring, comparing and analysing progress performance against progress benchmarks, such as actual start and completion dates, percentage of completed work and the remaining duration of current work.

118 Classification of costs

Project costs: the sum of the costs incurred throughout the project. Full life cycle costs: Total equity costs, i.e. total development, operational and maintenance costs. Fixed cost: Not affected by changes in business volume. Portable: subject to changes in production (fuel, raw materials, power). Direct costs: Costs directly attributable to the project (wages, bonuses). Indirect costs: Shared costs (rent, utilities). Sink cost: Daytime occurrence, irrecoverable.

119 Cost estimates

The cost estimates include a quantitative estimate of the costs that may be incurred to complete the project work, the contingency reserve to respond to identified risks and the management reserve for unplanned work. Cost estimates may be aggregated or detailed. Cost estimates should cover all resources used by the project. The estimate is a quantitative assessment of the cost of the project, ideally expressed as an area, and the size of the zone is predicted

120 Cost basis

The cost basis is the approved, time-based budget for the project, excluding any management reserve, which can be changed only through a formal change control process and is used as a basis for comparison with actual results. Cost benchmarks are generally prepared in time to summarize the estimated costs, usually in the S curve.

121 Comparison of tools to estimate costs

Estimation by analogy: cost savings, speed, roughness, usually more than used during project start-up and the pre-project period. Both analogies and parameter estimates use historical data, and analogies are an expert judgement. Parameter estimation: determined by both historical data and parameter models. Bottom-up estimates: the process of aggregation is the most accurate. Three point estimates: usually used when uncertainties are larger, and only when risks and uncertainties are uncertain.

122 Crude volume estimate for vs. certainty estimate

A rough scale estimate is a preliminary estimate made at the conceptual or start-up stage of the project without detailed data, and accuracy should be between 25 per cent and 75 per cent of the actual cost. A definitive estimate is an estimate obtained in the middle and later stages of the preparation phase of a project management plan and must be based on a detailed and complete bottom-up estimate of the WBS, with accuracy ranging from -5 to +10 per cent. Only definitive estimates can be used as a cost baseline for a project.

123 Cost basis vs. management reserve vs. contingency reserve

Management reserve: A portion of the project budget or project time is retained for management control purposes in addition to the performance measurement benchmarks. For unforeseen work within the scope of the project. Contingency reserve: the time or funds allocated to proactively respond to known risks within the progress or cost base. Cost basis: Approved project budgets allocated over time, excluding any management reserve, can be changed only through formal change control procedures, which can be used as a basis for comparison with actual results.

124 Profit analysis

Earned Value Analyse, EVA, compares real progress and cost performance with performance measurement benchmarks. EVA integrates scope, cost and progress benchmarks into performance measurement benchmarks. It is a way to assess project performance and progress by combining scope, progress and resource performance. EVA is a common performance measurement method that can take many forms.

125 Three indicators of earned value analysis

Plan value (PV): An approved budget is allocated for the planning of work. Earning value (EV): is the measure of the work done, and EV is used to calculate the percentage of the project completed. Actual costs (AC): The actual costs incurred in carrying out an activity within a given time period, AC has no ceiling and any costs incurred in achieving EV are calculated.

126 Planned value vs. earned on vs

Planned value: How much are we going to spend at some point in time, and the budget allocated to the planned work represents the earning value: the measurement of the work done represents the amount of work actually done. Actual cost: How much was actually spent. For example, I had planned to complete work worth 10 on 10 (planned value), and by 10 I found that I had done 15 (actual amount, earned value), so I was ahead of schedule.

127 Offset Measurement

Progress deviation (SV): Progress deviation is an indicator of project progress performance, calculated as SV = EV – PV. (b) When SV < 0, progress is delayed; Cost deviation (CV): Cost deviation is an indicator of project cost performance calculated as CV = EV-AC. Negative CVs are generally irreversible. (c) When CV = 0, the actual cost of consumption is equal to the budgeted value.

128 Progress performance index

Progress Performance Index (PI): The PPI is an indicator for comparing the progress of completed projects with the planned progress, calculated as SPI = EV/PV. (b) When SPI > 1.0, indicate that the completed workload exceeds the planned workload;

129 Cost performance index

Cost Performance Index (CPI): The CPI is an indicator of the value and actual cost of completed work, calculated as CPI = EV/AC. (c) When CPI = 1.0, the cost incurred to date is equal to the budget cost;

130 Cost estimates vs. budgeting

Cost estimates are cost estimates for specific activities (or work packages). The main function of cost estimation is to determine the amount of cost required to complete the project. Budget formulation is the aggregation of cost estimates for various activities (or work packages) and its primary function is to establish cost benchmarks against which project performance can be monitored and controlled.

PMBOK-aligned terminology - print-friendly review edition

131 Control quality vs. management quality

Control of quality is the matching of deliverables with quality standards into verifiable deliverables, so what is being done is to examine the results; Management quality is implemented in accordance with quality management plans, the effect of which is to improve the likelihood of achieving quality objectives, and to identify ineffective processes and causes of poor quality; Management quality lies at its core in process improvement and confidence assurance. At the heart of quality control lies the quality of deliverables. A process of concern, a result of concern.

132 Planning quality management

Planning quality management is the process of identifying quality requirements and/or criteria for the project and its deliverables and describing in writing how the project will demonstrate compliance with quality requirements and/or standards. Its main role is to provide guidance and direction on how to manage and validate quality throughout the project. A basic criterion for modern quality management is that quality is planned, not checked.

133 PDCA

The PDCA (David Ring) is a quality improvement method proposed by the quality management expert, Dai Ming. The quality management process is divided into plans (Plan, setting guidelines and targets, defining activity plans), implementation (Do, achieving the content of the plan), inspection (Check, summarizing the results of the implementation plan, taking care of the results, identifying problems), action (Action, managing the results of the review, identifying successful experiences and promoting them as appropriate) and establishing a PDCA cycle.

134 Six Sigma

Motorola has developed a six-sigma management approach based on statistical principles, a client-centred, data-based management concept that pursues almost perfectness. The core of the six Sigma management, which is the inheritance and development of comprehensive quality management, is to optimize the organization's operational capacity through a set of statistical science-based methods to identify problems, analyse causes, improve optimization and control.

135 Rule vs. standard

The rules are more binding, but not necessarily just government rules. You're subject to objective conditions, like those given by the organizing company, which can be understood in the examination as a career environmental factor, which is not controlled. The standard, like the PMBOK guide on which our test materials are based, is just a project management standard that gives some recognized best practices that can be used as a reference, but you can't say we learned these things as rules.

136 Mass vs. Level

Failure to meet quality requirements is certainly a problem. Low grades are not necessarily a problem. Low-grade, high-quality products are important killers for many producers to occupy larger market shares. The project manager is responsible for balancing with the project management team to achieve the required level of quality and grade at the same time. If there were not enough costs to meet the established project requirements, the project could be downgraded (reduced project scope and reduced functionality), but not at the expense of quality.

137 Tool technology - quality costs

Quality costs are divided into prevention costs, assessment costs (evaluation costs) and impairment costs (failure costs). Quality costs are the total cost of all efforts to obtain products and are the sum of consistent and non-consistent costs. The quality cost of the project should be between 3 and 5 per cent of the total cost of the project. Quality cost techniques were used to estimate costs and plan quality management processes.

138 Cost of prevention vs. assessed cost vs. failure costs

Prevention costs: Prevention of associated costs associated with products, deliverables or poor quality of service of a given project. Assessment costs: Costs associated with assessing, measuring, auditing and testing products, deliverables or services for specific projects. Cost of failure (internal/external): costs associated with a product, deliverables or service inconsistent with the needs or expectations of the parties concerned.

139 Consistency costs vs. non-consistency costs

Consistency costs are used during the project to prevent failure, including prevention costs (training, process documentation, planning, equipment, etc.) and assessment costs (testing, design confirmation, inspection, evaluation, audit, etc.). The non-consistency costs are used to address the costs of failure during and after the project, including internal failure costs (repatriation, scrap, additional inventory) and external failure costs (liability recognition, product recall, debt, warranty, complaint processing, etc.).

140 Role and content of quality management plans

Impact: Describe how applicable policies, procedures and guidelines are implemented to achieve quality objectives. Content: 1 Quality standards adopted 2 Quality objectives of the project 3 Quality roles and responsibilities 4 Quality deliverables and processes requiring quality review 5 Quality control and quality management activities planned for the project 6 Quality tools used by the project 7 Quality tools used by the project 7 Quality documentation

141 Test vs

Tests are an organized and structured survey designed to provide objective information on the quality of the products or services tested according to the needs of the project. The purpose of the test is to identify errors, defects, loopholes or other problems of non-compliance in the product or service. Early testing helps to identify non-compliance and reduces the cost of repairing non-compliance components. Inspection is to check that the deliverables do not meet the criteria. Tests are process-specific and check results.

142 Quality measurement indicators

Quality measurement indicators: dedicated to describing the project or product properties and how the control quality process will validate conformity. Examples of quality measurement indicators: percentage of missions completed on time, cost performance as measured by CPI, failure rate, number of identified daily defects, total monthly stoppage time, error per code line, customer satisfaction score, and percentage of needs covered by the test plan (i.e., test coverage).

143 Quality control measurements

Quality control measurements result in a written record of the results of the quality control activities (recorded in a format defined in the quality management plan) used to analyse and assess whether the project process and the quality of deliverables met organizational standards or specific requirements. Quality control measurements also help to analyse the process of producing these measurements to determine the correctness of the actual measurements.

144 Quality reports

Quality reports are used to report on quality management issues, recommendations for corrective measures (including re-entry, gap/facility remediation, 100 per cent inspection, etc.) and other findings in quality control activities. They may also include recommendations for process, project and product improvements. Quality reports may be graphic, data or qualitative documents containing information that can assist other processes and departments in taking corrective measures to meet project quality expectations.

145 Correlation

The causal chart, also known as the Ishikawa map, the "why-why analysis map", the fish gill map or the fish bone map, shows intuitively how the factors relate to potential problems or results. The use of cause and effect maps, which can be found at the back end of the product, can be traced back to responsible production behaviour, identifying the causes of quality from the source of production, and truly achieving quality improvement and improvement.

146 Analysis of the root causes of causal chart vs

Fish bone maps, also known as "cause and effect maps", "why-why analysis maps" and "Ishikawa maps", break down the causes of the presentation into discrete branches and help to identify the main or underlying causes of the problem. Analysis of root causes, focusing on the main causes of identification problems. Both the causal chart and the underlying cause analysis are the reasons for the problem. The difference is that the causal chart is only for reasons; the underlying cause is analysed for reasons for which recommendations are made to eliminate the problem and avoid re-issuing.

147 Flowchart vs. Control Chart

Flowchart: Graphical representation of a process to show the interrelationship between steps in the process. It can help improve the process and identify possible quality problems or possible quality checks. Control chart: used to determine whether a process is stable or predictable. Seven-point rule, seven consecutive points exceeding (or below) the average, is deemed to be out of control; exceeds the control limit, indicating loss of control; exceeds the specification limit, indicating unsatisfactory.

148 Histogram vs. Paretto

The histogram is a barchart showing digital data that shows the number of deficiencies per deliverables, the ranking of the causes of the deficiencies, the number of non-compliances of the processes, or other manifestations of project or product deficiencies. PARETO is a special histogram, organized by frequency of occurrence, showing how many defects each of the identified causes leads to, respectively, the purpose of the ranking is to focus on corrective measures. Project teams first address the causes of the most deficiencies.

149 Fragmentation vs. verification form

The scatterchart is a two-point relationship between variable and variable. One axis denotes any element of the process, environment or activity, and the other an axis denotes a quality defect. The verification form, also known as the Counting Table, is used to rationalize the ranking of matters in order to effectively collect useful data on potential quality issues. When an inspection is carried out to identify defects, a verification form is used to collect data on properties, such as data on the number of defects or the consequences.

150 Quality audit

Audit is a structured and independent process for determining whether project activities follow organizational and project policies, processes and procedures. Quality audits are usually performed by teams outside the project, such as the internal audit office of the organization, the project management office (PMO) or the external auditor of the organization. Quality audits may also confirm the implementation of approved requests for change (including updates, corrective measures, remedy for deficiencies and preventive measures).

151 Test vs. for vs. audit

Testing and inspection are tools and techniques for quality control. Audit is a tool technique for managing quality. The purpose of the test was to identify errors or deficiencies in the product or service; the inspection was to check that deliverables did not meet the criteria. If quality fails, it should be checked out. The audit primarily reviewed whether project activities were in compliance with organizational and project policies.

152 Problem resolution

Problem resolution finds solutions to problems or challenges. This includes collecting other information, critical thinking, creative, quantitative and/or logical solutions. Effective and systematic problem-solving is an essential element of quality assurance and quality improvement. Problems may be identified in quality control processes or quality audits and may be related to processes or deliverables

153 Content of problem resolution

Problem resolution finds solutions to problems or challenges. The use of structured problem solutions helps to eliminate problems and develop durable and effective solutions. The problem solution usually includes the following elements:

154 Tool techniques - statistical sampling

Statistical sampling refers to the selection of selected samples from the target population for inspection (e.g., eight randomly drawn from 198 products) only during quality control. The frequency and size of the sample should be determined in the planning quality process so that the quantity tested and the expected waste are taken into account in the quality cost. Statistical sampling has a rich knowledge system, and attention needs to be paid to distinguishing between attributes and variables.

155 Property sample vs. variable sample

Attribute sampling is the testing of one or more properties of a product. The results of the attribute check are "yes or no", "continue or do not continue", "deficient or non-deficient", "within or beyond tolerance", "correct or incorrect" etc. The sample of variables is the ability of a process to change by being measured and recorded. These variables may be minutes (time), degrees (temperature), kg (weight)

156 Resource calendar vs. organizational breakdown structure

The resource calendar specifies when and how long specific project resources will be available during the project period. A resource calendar can be established at the activity or project level. Additional resource attributes, such as experience and/or skill levels, origin and availability, need to be considered. This is all to be considered in the allocation of tasks. The organizational breakdown structure is intended for all parts of the organization as a whole, such as the procurement department, which is responsible for procurement and operations, rather than for the role of each individual within your team.

157 Resource structure vs. Resource breakdown structure

Based on the identification of critical skills and types of resources required for the project, the project management team constructs the project ' s resource structure based on the characteristics of the project and the actual needs of the project, as well as the identified roles, responsibilities and reporting lines of the project. Resource disaggregation structures are presented by category and type of resource. The resource category includes manpower, materials, equipment and supplies, and the type of resource includes skill levels, grade levels or other types applicable to the project.

158 Responsibilities distribution matrix

The matrix shows the distribution of tasks in the work packages for project resources. An example of a matrix is the Allocation of Duties Matrix (RAM), which shows the project resources allocated to each work package to illustrate the relationship between the work package or activity and the members of the project team. The matrix reflects all activities related to each individual, as well as all those associated with each activity, and it also ensures that one person is responsible for any given task, thus avoiding ambiguity.

159 RAM&RACI

The role allocation matrix (RAM), which shows the project resources allocated to each work package, is used to explain the relationship between the work package or activity and the members of the project team. An example of RAM is the RACI (implementation, accountability, consultation and information) matrix. If the team is composed of internal and external members, the RACI matrix is particularly useful for a clear delineation of roles and responsibilities.

160 Resource management plan

The resource management plan is a guide on how to classify, allocate, manage and release project resources. The resource management plan can be divided into a team management plan and a physical resource management plan, depending on the circumstances of the project. In general, the resource management plan should include the following elements: identification of resources, access to resources, roles and responsibilities, project organization chart, project team resource management, training, team building, resource control, and endorsement of the plan.

161 Team charter

Team charters refer to documents that document team values, consensus and work guidelines, and that specify the acceptable behaviour of project team members. (b) Communication guidelines; Team charters set clear expectations for acceptable behaviour by project team members. Early endorsement and adherence to clear rules would help to reduce misunderstanding and increase productivity.

162 Resource Calendar vs. Project Calendar

The resource calendar identifies working days, shifts, business hours, weekends and public holidays for each specific resource availability. During planning activities, information on potential available resources (e.g., team resources, equipment and materials) is used to estimate resource availability. The project calendar lists the working days or shifts in which the planned activities will be carried out, as well as the non-working days in which the planned activities will not be carried out. The project calendar affects all activities.

163 Access to Resource Tool Technologies - Negotiation

Interpersonal relations and team skills include, but are not limited to, negotiation. Many projects need to be negotiated against resource requirements and the project management team needs to negotiate with functional managers, other project management teams in the Executive Group and external organizations and vendors. (b) Key words: "Access to (replacement) resources, conflict in mandate allocation, consultations, etc.

164 Technology for accessing resource tools - pre-allocation

Pre-allocation refers to the pre-determined physical or team resources of the project. (b) The project depends on the expertise of the individual concerned;

165 Access to Resource Tool Technologies - Virtual Team

Virtual teams are a group of people who share common goals and carry out their respective responsibilities but have little or no time to work face to face. In a virtual team environment, communication planning becomes increasingly important. More time may be needed to set clear expectations, promote communication, develop conflict resolution, convene people in decision-making, understand cultural differences and share the joy of success. The virtual project team is based on the Internet and communications technology.

166 Form phase vs. shock phase

Forming phase: at this stage, team members know each other and understand the project and their formal roles and responsibilities in the project. At this stage, team members tend to be independent of each other, not necessarily honest. The shock phase: at this stage, the team began to work on the project, develop technical decision-making and discuss project management methodologies. If team members do not treat different views and opinions in a cooperative and open manner, the team environment may become counterproductive.

167 Normative stage vs. maturity phase

Normative phase: Team members start working together and align their work habits and behaviours to support teams, and team members learn to trust each other. (a) The maturity phase: after entering this phase, teams work as an organized unit, rely on each other among team members to resolve problems smoothly and efficiently. It is only when there are "dependency and efficient solutions" that we choose the stage of maturity, not just trust.

168 Dissolution phase

Dissolution phase: Team completes all work and team members leave the project. The release of personnel and the dissolution of the team are usually done after the deliverables of the project have been completed. Or break up the team in the end of the project or phase. In the five phases of the project team, the duration of a phase depends on team vitality, team size and team leadership. The project manager should have a better understanding of team dynamics to effectively lead the team through all stages.

169 Five Team Development Stages - Keywords

Forming: resources are added, new team members join, and members get to know one another. Storming: disagreement, conflict, controversy, and differences emerge. Norming: the team collaborates, adapts working habits and behavior, and starts building trust. Performing: the team cooperates and solves problems efficiently. Adjourning: the team is released and personnel are reassigned.

170 Team-building tool technology - centralized

Centralized office refers to the placement of many or all of the most active project team members in the same physical location to enhance the team ' s ability to work. Centralization can be either temporary or cross-cutting. The implementation of a centralized office strategy will allow for the use of team rooms, the posting of progress plans and other facilities that will enhance communication and collective awareness. Keywords: Increased efficiency, collaborative environment, improved communication, increased collective/team spirit.

171 Team-building tool technology -- communication technology

Communication techniques are critical in addressing team-building issues of centralized or virtual teams. It helps to create a harmonious environment for centralized office teams and promotes better mutual understanding among virtual teams, especially those whose members are dispersed in different time zones. Communication technologies available include shared portals, videoconferencing, audio conferences and e-mail/chat software.

172 Team-building tools -- interpersonal skills

Interpersonal and team skills include, but are not limited to: conflict management; influence; motivation; negotiation; Team-building: building an active working environment by organizing events to strengthen the social relationships of the team. Teambuilding activities aim to help team members work together more effectively. Informal communication and activities help to build trust and good working relationships. Team-building is essential in the early stages of the project, but it is more a continuous process.

173 Team Building Tool Technologies - Training

Training includes all activities aimed at enhancing the capacity of project team members, both formal and informal, through classroom training, online training, computer-assisted training, on-the-job training (provided by other project team members), coaching and training. Where project team members lack the necessary managerial or technical skills, the development of such skills can be part of the project.

174 Maslow's demand-level theory

Physical needs (food, water, air, clothing, etc.), safety needs (security, stability, protection from harm), social needs (families, belonging, friends), respect for needs (achievement, respect, drawing attention to others), self-realization needs (learning, development). Higher levels of demand can only be pursued if the lower levels of demand are met.

175 McGregor's X and Y theories

The X and Y theories advanced by McGregor are management theories about the motivation of people to work. It's a set of theories based on two totally opposite assumptions: the X theory, which considers people to be passive, lazy, trying to avoid work, lacking ambition and responsibilities; the Y theory, which considers people to be positive, willing to work, willing to progress, willing to take responsibility. Traditional management is more in favour of X theory, and modern management is more in favour of Y theory.

176 Herzberg's double-factor theory

Health factors and incentives. The former is a source of dissatisfiedness, and failure to do so undermines incentives, while good work does not raise incentives, such as working conditions, wages, relations between colleagues, security, jobs, etc., which are equivalent to lower levels of demand in the Maslowian theory, while the latter is a factor leading to satisfaction that can be truly stimulating, such as responsibility, self-realization, career development, recognition, etc., which is equivalent to higher levels of demand in the Maslowian theory.

177 Frum's expectation theory

The theory of expectation put forward by Vroom points out that the intensity of a behavioral tendency depends on the individual ' s expectations of the possible consequences of such behaviour and the attractiveness of the outcome to the individual. If a person believes that hard work will lead to successful results and that success will be matched by rewards, he is motivated to work hard.

178 McLelan's theory of motive for achievement

It is also called three needs to be addressed: the need for achievement, the need for power and the need for kinship. Managers should develop incentives based on the need for greater personal attention, such as challenging but achievable targets for those who need them, a more representative work environment for those who need power, and a cooperative rather than competitive work environment for those who need it.

179 Conflict management

In the project environment, conflicts are inevitable. Sources of conflict include scarcity of resources, prioritization of progress and differences in individual work styles. The use of basic team rules, team norms and mature project management practices (e.g. communication planning and role definition) can reduce the number of conflicts. Successful conflict management improves productivity and working relationships. At the same time, if properly managed, differences of opinion contribute to greater creativity and better decision-making.

180 Conflict management

Conflict management, if properly managed, has a difference of opinion in favour of greater creativity and better decision-making. If differences of opinion become negative, they should be addressed first and foremost by the members of the project team; if the conflict escalates, the project manager should assist in facilitating a satisfactory solution, using a direct and cooperative approach and dealing with the conflict as soon as possible and usually in private. If destructive conflicts persist, formal procedures, including disciplinary measures, can be used.

181 Causes of conflict

Scarcity of resources: resulting in differing views on resource allocation. Progress plan: There are conflicting views on the timing of project tasks. Prioritization of work: Different views were expressed on the prioritization of project work. Technical perspectives differ: views on technical issues are not uniform. Work style and management program: People work differently and prefer management programs. Cost: There is disagreement about how much should be done. Personality: the difference between a person and an individual.

182 Pull back the vs. and ease the vs. compromise

Withdrawal/avoidance: exit from actual or potential conflict, delay the issue until it is fully prepared, or push the issue to other personnel. Mitigation/inclusion: emphasis on coherence rather than difference; convergence and differences, taking into account the needs of others in order to maintain harmony and relationships. Compromise/conciliation: The search for a solution that would satisfy the parties to a certain degree of satisfaction for the temporary or partial resolution of the conflict sometimes leads to a "win-win" situation.

183 Force vs. to cooperate

Coercion/order: Promote one party ' s point of view at the expense of others; offer only win-lose solutions. Often power is used to impose urgent problems, an approach that usually leads to a "win or lose" situation. Cooperation/problems: a combination of different perspectives and opinions, a cooperative approach and open dialogue that leads to consensus and commitment can lead to win-win situations that are fundamental to problem resolution and can lead to long-term solutions.

184 Compromise vs. détente

The greatest difference between compromise and détente is that compromise deals with the conflict itself. Mitigation does not address the conflict itself. Compromise is the conflict itself, one step back. Mitigation is about emphasizing unity and diversity, or the absence of conflict management. Detachment is only a partial settlement of conflicts, and the gaps avoided remain objective. Compromise, on the other hand, can bring about a permanent and complete settlement of the conflict, all of which will be permanent as long as the compromise is not or will not be reversed.

185 The minority obeys the majority -- cooperate

A minority is subject to a majority vote, similar to a vote in a decision-making technique, which may include the following methods of decision-making: unanimity, majority consent or relative majority principle. Most of the agreement was to consult with the majority, to work together to solve the problem, not to use power to impose it. It is a convergence of views and opinions that leads to a solution that is accepted and committed by the majority. Thus, the majority is subject to cooperative settlement.

186 Relationships and team skills -- emotional

This refers to the ability to recognize, assess and manage personal, other and group emotions. The project management team is able to use intelligence to understand, assess and control the emotions of project team members, predict the behaviour of team members, identify team members ' concerns and track team members in order to reduce pressure and enhance cooperation.

187 vs. leadership

Impact: In a matrix environment, project managers usually have little or no command authority over team members, so that they influence the ability of the parties concerned in a timely manner is critical to the success of the project. Leadership: Successful projects require strong leadership skills, leadership being the ability to lead teams and inspire them to do their work. It encompasses a variety of skills, capabilities and actions. Lord

188 Performance review vs. trend analysis

Performance reviews are the measurement, comparison and analysis of differences in the use of planned and actual resources. Analysis of cost and progress work performance information could help to identify issues that could affect the use of resources. Trend analysis: In the course of project progress, the project team may use trend analysis to determine the resources required for future project phases based on current work performance information. The trend analysis examines project performance over time and can be used to determine whether performance is improving or deteriorating.

189 Steps to address the problem

Project managers should take orderly steps to address the problem, including: Identification of problems: Identification of problems; Definition of problems: Disaggregation of problems into manageable small issues; Survey: Data collection; Analysis: Identification of the root causes of the problem; Resolution: Selection of the most appropriate of the many solutions; Checking solutions: Identification of whether the problem has been solved.

190 Project communication management

Project communication management includes the processes required to ensure that project information is generated, collected, published, stored, accessed and ultimately disposed of in a timely and appropriate manner. Most of the time of the project manager (generally 75-90 per cent) is spent on communicating with team members and other stakeholders, whether they come from within the organization (at various levels of the organization) or from within the organization.

191 Planning communication for vs. management communication for vs. monitoring gap

Planning communication management: the process of developing appropriate methodologies and plans for project communication activities based on the information needs of each relevant party or group of related parties, available organizational process assets and the needs of specific projects

192 Formula for calculation of the number of channels of communication

The project manager should also use the number of potential channels or routes of communication to reflect the complexity of project communication. The channel of communication is the number of arranged lines of communication in the project, which looks like the number of telephone lines connecting all participants, and the formula for calculating the number of channels of communication if: $CC = nx(n-1)/2$ in which CC indicates the channel of communication and n indicates the number of members in the project.

193 Basic information on communication

Communication refers to the exchange of information, whether intentional or unintentional. The information exchanged may be ideas, instructions or emotions. (b) Formal (reports, memoranda, briefings) and informal (e-mails, improvisation);

194 Communications needs analysis

The communication needs analysis determines the information needs of project stakeholders, including the type and format of information, and the value of the information to the stakeholders. Project resources can only be used to communicate information that is conducive to success, or that can fail because of lack of communication. In planning project communication, it is necessary to identify and limit who should communicate with whom and who will receive what information. Sending the right information in the right way to the right person at the right time to do the right thing.

195 Communication model

Communications models should include at least three components: the sender, the recipient, and communication models are often a circular process. The sender should carefully check the coding of the information, determine the method of sending the information (the communication method) and confirm that the information has been understood by the recipient. The recipient should decode the information carefully and ensure that it is properly understood.

196 Interactive communication vs. lated communication

Interactive communication: Multiple exchange of information between parties or multiple parties. This is the most effective way to ensure consensus among all participants on a topic (meetings, teleconferences, videoconferences).

197 Pull communication vs. push communication

Pull communication: used in situations where the amount of information is high or the audience is high. It requires the recipient to independently access the content of the information (Enterprise Innernet, e-Online Course, Knowledge Bank) Push-through communication: to send the information to a specific recipient who needs to know the information. This approach ensures the dissemination of information but does not ensure that it reaches or is understood by the target audience (letters, memos, reports, e-mails, faxes, voicemails, press releases).

198 Choice of communication methods

Interactive communication: 1, which needs to be agreed through multiple exchanges of information; 2, which generally applies to more important content and information interaction; Push communication: 1, ensure that information is published, but does not ensure that it reaches the target audience, or that it is understood by the target audience: 1, multiple target audiences " take what is needed " for multiple messages.

199 Content of communication management plan

5W (who, when, what, how, where); Communication needs of stakeholders; Information to be communicated, including language, format, content, level of detail; Reasons for publication of relevant information; Time and frequency of publication of information and communication of receipt or response; Persons responsible for communication of relevant information; Persons or groups to be received of information; Technology or methods for transmission of information; Problem upgrading process; Guides and templates for meetings and e-mails, etc.

200 Definition and role of communication management plans

Definitions: As the main output of the planning communication management process, the communication management plan is an important document to guide project communication and is part of the project management plan. The plan describes how project communication will be planned, managed and monitored. Impact: Ensure that appropriate information is communicated to stakeholders in various forms and means. Sending the right information in the right way to the right person at the right time to do the right thing.

201 Participation of stakeholders in the vs. communication management plan

The process of developing a methodology for project-related participation based on the needs, expectations, interests and potential impact on the project. The focus is on demand horizons, similar to PR schemes. Communication management plan: The related party management plan is followed by a part of the participation plan, but focuses more on communication, such as the transmission of information.

202 Relationship skills -- active listening

Interact with speakers and summarize the dialogue to ensure effective information exchange. The following process uses active listening skills: Management of project knowledge: Active listening helps to reduce misunderstandings and facilitates communication and knowledge sharing. Management communication: Active listening technology includes informing of receipt, clarifying and confirming information, understanding and removing barriers to understanding. Monitoring participation of stakeholders: by actively listening, reducing errors in understanding and communication.

203 Communication style assessment

Communication style assessments are interpersonal and team skills. Communication style assessments are a technique used in planning communication management to identify preferred communication methods, formats and content for communication with stakeholders. Frequently used for parties that do not support the project. An assessment of the participation of stakeholders could be undertaken, followed by a communication style assessment.

204 Interpersonal relationship skills -- conference management

Conference management is taking steps to ensure that meetings meet their expected objectives effectively and efficiently, including preparing the agenda, ensuring that representatives of each key stakeholder group are invited, and preparing and transmitting follow-up minutes and action plans. Conference management skills were used in three processes, namely, project charter development, project management plan development and management communication.

205 Conference management steps

The following steps should be taken in planning the conference: (1) preparation and publication of the agenda of the meeting (which includes the objectives of the meeting); (2) ensuring that the meeting begins and ends at the specified time; (3) ensuring that appropriate participants are invited and present; (4) relevance; (5) addressing expectations, issues and conflicts in the meeting; and (6) recording all actions and those assigned to them.

206 Project reporting

The project reports collect and publish work performance information, including status reports, progress measurements and projections. The project management team should regularly collect, compare and analyse baseline and actual data to inform and communicate project progress and performance and project results. Reporting on performance requires that information be provided to each interested party in an appropriate manner. The format of work performance reports can range from simple status reports to detailed narrative reports.

207 Status report vs. progress report

The status report describes the project phase of the project at a given point in time. The status report describes the status of the project in terms of scope, time, cost, quality objectives and quantitative data. The progress report describes the completion of the project team ' s work at a given time frame. Project progress reports are the simplest written reports available on the monitoring, inspection and status of project progress and trends.

208 Analysis of expected monetary value (EMV)

EMV is a statistical method for calculating average results (analysis under uncertainty) when certain situations may or may not occur in the future. The EMV for opportunity is usually positive, while the EMV for threat is negative. EMV is based on a risk-neutral assumption that it is neither safe nor risky. By multiplying the value of each possible result with its probability of occurrence, and adding all the multipliers, the EMV of the project can be calculated. This technology is often used in decision tree analysis.

209 Risk management overview

The four elements of risk are event (condition), cause, probability and consequence. What kind of event is the risk (condition), what causes it, what is the probability that it will occur, what consequences it will have? The purpose of risk management is to minimize the negative impact of risk on project objectives, seize the opportunities presented by risk and increase the benefits for project stakeholders.

210 purely risk vs. speculative risk

By classifying risks according to their consequences, they can be classified as purely risk and speculative. Pure risk does not bring opportunities and benefits. There are only two possible consequences of pure risk, i.e. to create and not to cause loss, which is a loss to society as a whole from which no one benefits; and speculative risk, which can bring opportunity, gain benefits and imply threats and losses.

211 Known risks of vs. predictable vs. unpredictable

Risks can be classified according to their predictability and can be classified as known, foreseeable and unpredictable. The known risks are those that can be identified and the consequences foreseeable. Risks are known to be highly probable, but with minor consequences; predictable risks are those that can be foreseen in the light of experience, but whose consequences are not foreseeable. (b) Unpredictable risk is defined as an unforeseen risk, also known as an unknown, unidentified risk.

212 Project vs. Technology vs. Business Risk

At the macro level, risk can be classified as project risk, technical risk and commercial risk. Project risks refer to potential budget, progress, personnel, resources, user and demand issues, and their impact on projects. Technical risks refer to potential design, realization, interface, testing and maintenance issues. Technical risks threaten the quality of the project and the scheduled delivery time. Commercial risk refers to operational risk at the organizational level, which threatens the viability of the product.

213 Known-known risks

Known - Known risks: The risks have been identified and analysed, and they are known about the risks and the likelihood and consequences of their occurrence, and are usually factored into the costs of specific projects at the calculated level of risk. In other words, for known-known risks, the costs (at expected monetary value) are charged directly to the project activities.

214 Known - unknown risk

Known-unknown risks: Risks that have been identified but for which the probability or consequences are not clear can normally be met from emergency reserves (including contingency time and funds). For known-unknown wind risks, the potential cost is to be included in the project ' s contingency reserve, which is part of the project cost baseline.

215 Unknown - Unknown risk

Unknown - unknown risk: a completely unknown risk that has never been encountered in the past. For example, SARS was a risk prior to the first case of atypical pneumonia. If it happens, it's managed by a reserve. The use of the contingency reserve requires approval by the project manager and management approval for the utilization of the management reserve. As a result, the management of unknown-unknown risks is not normally the responsibility of the project manager

216 Residual risk

Residual risk (Residual Risk) is also referred to as residual risk and is a risk that remains after a risk response, that is, a strategic risk that is not controlled by the project team and a process risk for the business processes. Residual risks are usually acceptable. If residual risk exceeds the acceptable level of risk in the organization, the project team must arrange further risk response measures to reduce residual risk to an acceptable level.

217 Residual risk vs. secondary risk

Residual risks are those that remain after the risk response, that is, those strategic risks and process risks that are not controlled by the project team. Secondary risk, also referred to as secondary risk, is a risk associated with addressing a risk. If the former risk is not addressed, the latter risk (secondary risk) does not occur. For secondary risks, the project team should identify and plan responses.

218 Single & corporate project risk

Individual project risk: Uncertain events or conditions that, if they occur, have a positive or negative impact on one or more project objectives. The management of individual project risks is designed to take advantage of or reinforce positive risks (opportunities) and avoid or mitigate negative risks (threats). Corporate project risk: The impact of uncertainty on the project as a whole is a positive and negative variation in project outcomes faced by the parties concerned. It stems from all the uncertainties, including individual risks.

219 Planning risk management vs. risk identification

Planning risk management is the process of defining how to implement project risk management activities, and its main role is to ensure that the level, methodology and visibility of risk management are matched with the level of project risk and the importance of the project to the organization and other stakeholders. Risk identification is the process of identifying individual project risks, as well as the source of overall project risks, and documenting risk characteristics.

220 Risk qualitative analysis vs. risk quantitative analysis

Qualitative risk analysis provides the basis for subsequent analysis or action by prioritizing risks by assessing the probability and impact of individual project risks and other characteristics. Quantitative risk analysis is the process of quantitative analysis of the combined impact of the overall project objective on identified individual project risks and other sources of uncertainty.

221 Planning vs. implementation risks to address vs. oversight risks

Planning risk response: the process of developing options, selecting coping strategies and agreeing on response actions to address overall project risk exposure and individual project risks. Implementing risk response: the process of implementing an agreed risk response plan. Oversight risks: Overseeing the implementation of agreed risk response plans, tracking identified risks, identifying and analysing new risks and assessing the effectiveness of risk management throughout the project.

222 Risk management plan

The risk management plan describes how project risk management is organized and implemented and is a sub-plan for the project management plan. The risk management plan includes risk management strategies, methodology, roles and responsibilities, funding, timing, risk categories, related party risk preferences, risk probability and impact definitions, probability and impact matrix, reporting format, and tracking.

223 Risk register

Risk register: The risk register records detailed information on identified individual project risks. The results of these processes are also recorded in the risk register as a result of the implementation of the qualitative risk analysis, the planning of risk response, the implementation of risk response and the monitoring of risk. Depending on the specific project variable (e.g. size and complexity), the risk register may contain limited or extensive risk information.

224 Risk report

The risk report provides information on overall project risks, as well as summary information on identified individual project risks. In the project risk management process, the preparation of risk reports is a gradual process. Risk reporting may include (but not be limited to) sources of overall project risk. Describe the most important drivers of overall project risk exposure. Summary information on identified individual project risks.

225 Probability and impact matrix

Probability and impact matrices are tools and techniques for risk qualitative risk analysis, risk probability refers to the likelihood that the risk will occur, while risk impact refers to the impact of the risk upon the project objective once it has occurred. These two elements of risk relate to specific risk events rather than to the project as a whole. To ensure the quality and credibility of the qualitative risk analysis process, different levels of risk probability and impact need to be defined. Risks are prioritized according to their potential impact on the achievement of project objectives.

226 Sensitivity analysis

Sensitivity analysis techniques are used only for quantitative risk analysis. Sensitivity analysis, also known as sensitivity analysis, helps to determine which risks have the greatest potential impact on the project. Stabilize all other uncertainties in the baseline, and then look at the extent to which each factor changes will affect the target. The common expression of sensitivity analysis is tornado maps, which are used to compare the relative importance and impact of highly uncertain variables with those of relative stability.

227 Impact Chart

Impact mapping techniques are used only for quantitative risk analysis. The impact diagram is a graphical representation of causality, event sequence and other relationships between variables and outcomes. Impact maps are graphic aids for decision-making under uncertain conditions. It manifests a situation in a project or project in a series of entities, outcomes and impacts, as well as in their relationships and interactions.

228 SWOT analysis

It is a case-by-case examination of the strengths, weaknesses, opportunities and threats of the project (SWOT). When identifying risks, it will include internal risks, thus widening the range of risks identified. First, focus on the project, organization or general area of operation, identifying the strengths and weaknesses of the organization; then identify the opportunities that organizational advantages can bring to the project and the threats that organizational disadvantages may pose. The extent to which organizational advantages can overcome threats could also be analysed.

229 Quantitative Risk Analysis Tool Technology - Simulation

The simulation is intended to use a model to calculate the potential impact of uncertainty on the details of the project on its objectives. Simulations usually use Monte Carlo technology. In simulations, multiple (repeated) calculations are performed using project models. For each calculation, randomly extracted values (such as cost estimates or activity duration) from the probability distribution of these variables are used as inputs. Multiple calculations result in a probability distribution histogram (e.g. total cost or completion date).

230 Quantitative Risk Analysis Tool Technology - Simulation

The simulation is intended to use a model to calculate the potential impact of uncertainty on the details of the project on its objectives. Simulations usually use Monte Carlo technology. For cost risk analysis, simulations are required using cost estimates; For progress risk analysis, simulations are required using progress network maps and duration estimates.

231 Risk orientation

Risk-driven: risk lover, risk preference, risk-taking. Prefer risk choices, their risk coping strategy preferences are accepted. Risk neutrality: Under the same expected return conditions, there is no preference for determining results and uncertain outcomes. Risk aversion: A conservative approach to risk, for which the preferred strategy for risk response is circumvention.

232 Five types of response to threats -- reporting

If the project team or project sponsor believes that a threat is outside the scope of the project or that the proposed response exceeds the authority of the project manager, a reporting strategy should be used. The reported risks will be managed at the project cluster level, at the portfolio level or at other relevant parts of the organization, rather than at the project level. The project manager determines who should be notified of the threat and conveys detailed information about the threat to the person or organizational department.

233 The threat's five strategies -- circumvention

Change the project management plan to completely eliminate the threat. The project manager may also isolate the project objectives from the impact of risks or change the threatened objectives, such as extending progress, changing policies or narrowing the scope. The most extreme avoidance strategy is to cancel the entire project. Some of the risks that arise in the early stages of the project can be addressed by clarifying needs, obtaining information, improving communication or obtaining know-how

234 Five types of response to threats -- diversion

Transfer some or all of the negative effects of a certain risk to third parties, together with liability for the response. The risk is transferred by simply shifting the responsibility for risk management to the other side, rather than eliminating the risk. Risk-transfer policies are slightly more effective in addressing the financial consequences of risk. Using a risk transfer strategy, risk costs are almost always paid to the risk-bearer. Risk transfer includes insurance, performance bonds, guarantees and bonds. Contracts could be used to transfer certain specific risks to the other party.

235 Five types of response to the threat -- mitigation

Reduce the probability and/or impact of an adverse risk event to an acceptable threshold. Early action to reduce the probability of a risk occurring and/or its potential impact on the project is often much more effective than risk before seeking redress. Examples of mitigation measures include the use of less complex processes, more testing or the selection of more stable suppliers.

236 Five types of response to the threat -- yes

The project team has decided not to change the project management plan to address a risk or to find any other reasonable response strategy. The strategy can be passive or proactive. Passive acceptance of wind risk requires only recording of this strategy and does not require any other action; it will be handled by the project team when the risk arises. The most common proactive strategy is to establish a contingency reserve with time, funds or resources to deal with risks.

237 Five coping strategies for opportunities - reporting

If the project team or project sponsor believes that an opportunity is outside the scope of the project or that the proposed response is beyond the authority of the project manager, the reporting strategy should be used. The reported opportunities will be managed at the project cluster level, at the portfolio level or at other relevant parts of the organization, rather than at the project level. The project manager determines who should be notified of the opportunity and communicates detailed information about the opportunity to the person or organizational unit.

238 The five coping strategies of the opportunity - open

If the organization wants to ensure that opportunities are realized, it can use its own strategy for risks that have a positive impact. This strategy aims to eliminate uncertainty associated with a particular positive risk and to ensure that opportunities do occur. Direct opening includes allocating the most capable resources of the organization to projects to reduce completion time or cost savings.

239 Opportunity Policy - Sharing vs

Sharing: Allocating some or all of the responsibility for the opportunity to respond to it to third parties best able to seize it for the benefit of the project, including establishing risk-sharing partnerships and teams, and forming companies or consortia for special purposes, with the aim of taking full advantage of the opportunity to benefit all parties. Accepted: Be happy to use opportunities when they occur, but not to pursue them.

240 Opportunity Policy - Improved vs

Improved: Increased probability and/or positive impact of opportunities. Identifying and maximizing the key factors that affect the occurrence of positive risks increases the probability of opportunities occurring. Examples of improved opportunities include increased resources for early completion of activities. Accepted: Be happy to use opportunities when they occur, but not to pursue them.

241 Open vs. Upgrade

A pioneering approach aimed at removing uncertainty associated with a particular positive risk is to ensure that 100 per cent of opportunities occur. Improvement is a positive impact in terms of increasing the probability of opportunities occurring. (b) Increased resources include early completion of activities.

242 Post-risk response

If the risk occurs, the risk register is first consulted. 1. If documented, implement the response measures recorded therein; and, if the risk is not documented and does not lead to consequences, re-identify the risk, conduct risk analysis, plan for a more visible response, update the risk register and implement the established risk response

243 Emergency Plan vs. Bombback

Contingency plan: Pre-developed risk response plan. A pre-determined response to be taken by the project team when a potential risk event actually occurs A "Backback" plan: a backup response plan for a particular risk to be activated if the primary response plan (that is, the usual contingency plan) does not work. Implements the selected strategy when it is ineffective or when there is an accepted risk. Both the ejection plan and the contingency plan can target threats or opportunities.

244 Contingent Response Strategy - Workaround

A workaround is an unplanned response taken when an unidentified or unanalyzed threat occurs during risk monitoring. It is an urgent response to a negative risk that has already happened. Fallback plans and contingency plans may address threats or opportunities, but workarounds apply only to threats that require immediate action.

245 Risk audit

Through risk audits, examine and document the effectiveness of risk responses in addressing identified risks and their root causes, as well as the effectiveness of the risk management process. There can be either a risk audit in a daily project review or a separate risk audit meeting. The format and objectives of the audit should be clearly defined before the audit is implemented.

246 Risk re-evaluation

Surveillance risks often require the identification of new risks, the re-evaluation of existing risks and the removal of obsolete risks, as well as the removal of non-existent risks and the release of corresponding stockpiles. Project risk re-assessment should be scheduled. The number and level of detail of repeated re-evaluations should be based on the progress of the project objectives.

247 Risk vs. problem

Risk is an uncertain event or condition that, if it occurs, affects at least one project objective. Positive is opportunity, negative is the threat. The risk register records the results of risk analysis and risk response planning, and first documents are required to update the risk profile when it changes. The problem was a real one, and it would be a problem to be careful that risks arose and become projects. Issue logs are used to document and monitor the resolution of problems.

248 Risk-related sequence

Risk-related sequence: Identification of risk - > Qualitative analysis of risk - > Quantification of risk (not necessarily all risks need to be quantified) - > Plan risk response - > Risk implementation should occur.

249 Planning for the procurement of vs. to implement vs. control acquisition

Planning procurement management: is the process of recording project procurement decisions, clarifying procurement methods and identifying potential vendors. Conduct procurement: is the process of obtaining the seller ' s response, selecting the seller and awarding the contract. Procurement control: is the process of managing procurement relationships, monitoring contract performance, implementing necessary changes and corrections, and closing contracts.

250 Fixed fixed-price contract (FFP)

FFP is the most commonly used type of contract. Most buyers prefer such contracts because the price of the procurement is fixed at the outset and changes are not permitted (unless the scope of work changes). Any increase in costs due to poor performance of the contract is the responsibility of the seller. Scope of use: Under the FFP contract, the buyer must precisely define the products and services to be procured, and any change in procurement norms may increase the cost to the buyer.

251 Total price plus incentive fee contract (FPIF)

Some deviations in performance are allowed and financial incentives are given to achieve the stated objectives. Often, incentives relate to the costs, progress or technical performance of the seller. Performance targets are set at the outset, and the final contract price is determined on the basis of the seller ' s performance after the completion of all work. Scope of application: Some flexibility is provided for both the buyer and the seller. In the FPIF contract, to set a maximum price, the seller must complete the work and bear the full cost above the upper limit

252 Total plus economic price adjustment contract (FP-EPA)

The final adjustment of the contract price in a predetermined manner is permitted on the basis of a change in conditions (e.g. inflation, increase or decrease in the cost of certain special commodities). The EPA clause must provide for reliable financial indicators to be used for the definitive adjustment of the final price. Scope of application: performance by the seller extends over a considerable period of time (number of years) or multiple long-term relationships between the seller and the buyer. Protection of both parties from situations beyond their control.

253 Cost plus fixed-cost contract (CPFF)

To reimburse the seller for all available costs incurred in performing the work under the contract and to pay the seller a fixed fee calculated as a percentage of the initial cost of the project. The costs can only be paid for work done and do not vary depending on the seller ' s performance. Unless the scope of the project changes, the cost amounts remain unchanged. Scope of application: The buyer cannot precisely define the products and services to be procured and the seller ' s performance does not affect the seller.

254 Cost plus incentive costs (CPIF)

The seller reimbursed all available costs incurred in performing the work under the contract and paid the seller a pre-determined incentive fee when the seller met the performance objectives set out in the contract. In the CPIF contract, where the final cost is lower or higher than the original estimated cost, the buyer and seller are required to share the savings or share the excess in accordance with the pre-agreed cost-sharing ratio. Scope of application: The buyer is unable to define the products and services to be procured precisely by controlling the seller ' s performance and sharing opportunities and risks.

255 Cost plus incentive (CPAF)

All legal costs incurred in performing the contract were reimbursed to the seller, but a substantial portion of the costs could be paid to the seller only if certain general and subjective performance criteria specified in the contract were met. It is entirely up to the buyer to determine the award based on its own subjective judgement of the seller ' s performance, and the seller is usually not entitled to complain. Scope of application: The buyer cannot precisely define the products and services to be procured and the buyer is subjective in its performance.

256 Engineering contract (T&M)

The works contract (also referred to as the time and materials contract) is a mixed contract with the characteristics of a cost-reimbursable contract and a fixed-price contract. Such contracts often apply to the expansion of personnel, the hiring of experts or the seeking of external support where accurate job descriptions cannot be produced quickly. Several conditions of the contract: short-term, immediate start, target-setting, but uncertain workload.

257 Contract selection - terms of reference

Analysis of clarity of scope of work: 1. Use of the total price contract if the scope of work is clear and the project design is already detailed. Where the nature of the work is clear, but the scope is not clear and the work is not complex and contracts are concluded quickly, the use of the contract of works is made. Where the terms of reference are not clear, the cost reimbursement contract is used.

258 Contract selection - from a risk perspective

From a risk perspective: For buyers: the lowest risk is the fixed total price contract (FFP); the greatest risk is the cost plus fixed cost contract (CPFF). For sellers: the lowest risk is the cost plus fixed cost contract (CPFF); the greatest risk is the fixed fixed-price contract (FFP). Whether this determination is not to say that the risk is borne by the buyer is to say that the risk is the least for the buyer.

259 Homemade or outsourced analysis

Home-made or outsourced analysis is a generic management technique used to determine whether a particular job is best performed by the project team or must be procured externally. In some cases, while the project organization has the corresponding capacity within itself, procurement from outside the organization is also required to meet the demands of progress, as the related resources are being implemented in other projects. Homemade or outsourcing analyses should take into account all associated costs, including direct and indirect costs.

260 Content of procurement management plan

The type of contract to be used; Risk management matters; How to manage multiple suppliers; Standardized procurement documents (if required); How to guide sellers in preparing and maintaining WBS; How to identify pre-qualified sellers, if any; Constraints and assumptions that may affect the procurement process; Procurement measurement indicators for contract management and vendor evaluation; Whether independent estimates are required and whether independent estimates should be used as evaluation criteria; How to determine the lead time required for procurement.

PMBOK-aligned terminology - print-friendly review edition

261 Procurement statement of work

Procurement statement of work (SOW) defines only the project scope to be included in the relevant contract. The procurement of SOW should describe in detail the products, services or outcomes to be procured in order to enable potential sellers to determine whether they are capable of providing them. SOW may include specifications, required quantities, quality levels, acceptance methods and standards, work performance data, time limits, duty station, currency, payment schedule, guarantees and other elements.

262 Tender documents

The solicitation documents are used to solicit proposals from potential sellers. If the choice of the seller is based primarily on price (e.g., when a commercial or standard product is purchased), the term tender, tender or quotation is usually used; if other factors (e.g. technical capability or technical methods) are essential, the term " proposal " is usually used. The solicitation documents may be invitations to information, invitations to quotation, invitations to make recommendations or other appropriate procurement documents.

263 Supply-side selection criteria

The supplier selection criteria are usually part of the procurement documents. These criteria were developed to rate or rate the seller ' s proposal. The criteria may be objective or subjective. If the procurement is easily obtained from many qualified sellers, the selection criteria may be limited to the purchase price. Other selection criteria need to be identified and documented for more complex products, services or outcomes. Examples include: capacity and potential; delivery dates; technical expertise and methodologies; and financial stability of companies.

264 Weighted split of proposal evaluation techniques

The weighting method, by which tenders are rated according to the relevant evaluation criteria and the criteria are given a weight, and the weighting is aggregated to obtain the order of ranking of potential sellers. In practice, the criteria for scoring are as follows: based on objective facts; strict control of discretion; achievement of clear scores; implementation of national regulations reflecting national policies; and easy assessment of scoring criteria

265 Independent estimate of proposal evaluation techniques

The price of a potential seller is estimated independently, compared with the buyer's independent estimate prepared in advance (known as "bottom") and graded by difference. If there is a clear difference between the two, it may indicate that SOW is flawed (or unclear) and/or that a potential seller has misunderstood (or failed to respond fully) SOW.

266 Procurement negotiations

The negotiations should include liability, competence to make changes, applicable terms and laws, technical and commercial management methods, ownership, contract financing, technical solutions, overall progress plans, payments and prices. The negotiation process ends with the formation of a contract document that is enforceable by both the seller and the buyer, and the final wording of the contract should reflect the full agreement of the parties. The negotiations were aimed at mutual understanding and at protecting future relations between the parties after the establishment of the contract.

267 Implementation of procurement -- bidders ' conference

The bidder ' s meeting, also known as the contractor ' s meeting, the seller ' s meeting, the supplier ' s meeting or the pre-bid meeting, is a meeting between the buyer and all potential sellers before proposals or proposals are submitted. The purpose of the meeting was to ensure that all potential sellers had a clear and consistent understanding of the procurement, including technical and contractual requirements, and that no bidder would receive preferential treatment. Answers to questions should be included in the procurement document in the form of amendments.

268 Procurement -- inspection of vs. audits

With regard to procurement management, the relevant inspection and audit should be performed by the buyer in accordance with the terms of the contract and supported by the seller in the execution of the project. Inspections and audits can verify the extent to which the seller has complied with the contract in the course of its work or in terms of the deliverables it has accomplished. If the terms of the contract so permit, the buyer ' s procurement staff may be included in certain inspection and audit teams.

269 Procurement performance reviews

The procurement performance review is a structured review designed to examine the extent to which the seller has completed the project within the prescribed cost and schedule, and the quality requirements. It could include inspections conducted by the buyer, examination of relevant documents produced by the seller and quality audits conducted during the performance of the seller. Procurement performance reviews may be part of the project status review. The performance of key suppliers is usually taken into account in project status reviews.

270 Project delay claims

Claims for work-time delay: Compensable delay (the seller did not make a fault and allowed the seller to extend the work) and unforgivable delay (the seller made a mistake and the seller did not allow the extension), compensation for delay (the seller made no fault but the buyer made a mistake not only to allow the seller to extend the work period but also to compensate the seller for the economic loss suffered by the seller as a result of the delay), non-compensable delay, delay on critical routes and non-critical road delays, joint delay and non-joint delay.

271 vs. Change of adverse site conditions for vs

In addition to the claim for delay, there are three types of circumstances giving rise to the claim. (c) Accomplishment claim: a claim for increased costs resulting from the investment of additional human, material and financial resources in unit time. Change claim: Change of contract may be caused by a number of causes. Claims for adverse site conditions: The seller can usually claim from the buyer for losses arising from unforeseen adverse site conditions.

272 Causes and principles of the claim

An important prerequisite for a contractual claim is the existence of a breach of contract and of facts between one or both parties and the resulting loss for which the other party is liable. The seller may be granted an extension of the term of work in cases where the claim for the contract is for objective reasons and is unforeseeable to the buyer, e.g. exceptional abnormal weather meets the agreed terms of the contract for exceptional abnormal weather, but

273 Procurement documents vs. procurement documents

Procurement documents: The procurement documents are the outputs of the planning process and include only solicitation documents prepared by the buyer. Procurement documents: Procurement documents are the control of the output of the procurement process, including (but not limited to) procurement contracts and all supporting progress plans, unauthorized contract change requests and approved change requests. The procurement documents also include technical documents and other work performance information, such as deliverables, vendor work performance reports, guarantees, etc., prepared by the seller.

274 Content of Party Analysis

The relevant party analysis is the systematic collection and analysis of quantitative and qualitative information to determine who should be considered in the project, to identify the interests, expectations and impacts of the parties involved and to link them to the purpose of the project through the relevant party analysis. Relevant-party analysis also helps to understand the relationships between the parties involved in order to use them to build alliances and partnerships, thus increasing the likelihood of project success.

275 Relevant party analysis steps

Identify all potential project stakeholders and their relevant information, such as their roles, sectors, interests, level of knowledge, expectations and impact. Identify the potential impact or support provided by each of the parties concerned and group them in order to develop management strategies. Where there are many parties involved, the key parties must be sequenced.

276 Stakeholder Mapping Models

Power/interest grid: groups stakeholders by their authority and level of concern about project outcomes. Power/influence grid: groups stakeholders by authority and active involvement in the project. Influence/impact grid: groups stakeholders by their active involvement and their ability to change the project plan or execution approach.

277 Methodology for identifying party data performance

Power/interest: Grouping according to the power (power) of the party concerned and the degree of interest (interest) in the outcome of the project; Highlighting models: Categorizing the party concerned according to its power (capacity to impose its will), urgency (needing immediate attention) and legitimacy (right to participate), applicable to complex related community or intra-party relationships, and determining the relative importance of identified parties.

278 Power/interest

Powers/interests are grouped according to the size of the authority (authority) and the level of interest (interest) in project results. Monitoring of low-power and low-interest (minimization of effort), communication of low-power and high-interest, satisfaction of high-interest and high-interest needs Heavy

279 Participation of stakeholders in the assessment matrix

The stakeholder participation assessment matrix is used to compare current and expected levels of participation. The level of participation can be divided into: lack of knowledge, resistance, neutrality, support, leadership. The necessary communication should be carried out to effectively channel the participation of each interested party to the project, based on the current gap between the level of participation expected. Closing the current gap with the expected level of participation is an essential part of monitoring the involvement of stakeholders.

280 Identification of stakeholders involved in vs. planning

Identification of stakeholders: The process of periodically identifying project stakeholders, analysing and documenting their interests, participation, interdependence, impact and potential impact on project success. Planning stakeholder participation: The process of developing a methodology for project stakeholder participation based on the needs, expectations, interests and potential impact on the project.

281 Managing vs. monitoring party involvement

Manage stakeholder participation: communicate and collaborate with stakeholders to meet their needs and expectations, address issues and facilitate a process of reasonable participation by stakeholders. (c) Monitoring participation by stakeholders: monitoring participation by stakeholders is the process of monitoring the relationship between the parties involved in the project and leading to reasonable participation by stakeholders through the revision of the strategy and plan for participation. Its main role is to maintain or enhance the efficiency and effectiveness of participation of stakeholders as projects progress and environmental changes.

282 Participation of stakeholders in the plan

Information in the registry of the party concerned; The required level and current level of participation of the key interested party; The scope and impact of the change by the party concerned; Interrelation and potential cross-section of the party concerned; The communication needs of the party concerned at the current stage of the project; Information that needs to be distributed to the party concerned; The rationale for and possible implications for the participation of the party concerned; The time frame and frequency of dissemination of the required information to the party concerned; The methodology for updating and optimizing the relevant party management plan as the project progresses.

283 Related party management strategy

The related party management strategy sets out how to increase the support of the parties concerned and reduce their negative impacts throughout the life cycle of the project. It includes the following: the key parties that have a significant impact on the project; the extent to which each of the parties involved is expected to be involved in the project; the grouping of the parties concerned and the measures managed by the group; the frequent use of party analysis matrices to show the Party ' s management strategy; and the fact that some information related to the Party ' s management strategy may be too sensitive to be included in a publicly available document.

284 Register of stakeholders

The relevant party register is used to record all details of the identified party, including (but not limited to): Basic information: name, position, location, project role, contact information; Assessment information: main needs, main expectations, potential impact on the project, and which stage of the life cycle is most closely related; Relevant party classification: internal/external, supporter/neutral/opposers, etc.; Relevant party registers should be regularly viewed and updated, and stakeholders may change.

285 Stakeholder Engagement Plan vs. Communications Management Plan

The stakeholder register records stakeholder information, authority, influence, expectations, and needs. The stakeholder engagement plan builds on that information and focuses on management strategies based on stakeholder influence and needs. In exam terms, it is closer to a public-relations plan for stakeholder engagement.

286 Managing the expectations of stakeholders

Management stakeholders ' expectations relate to communication activities targeting project stakeholders to influence their expectations, address their concerns and resolve problems. By actively managing the expectations of the parties concerned, the risk of the project not meeting its goals and objectives owing to unresolved issues between the parties concerned could be reduced, as well as confusion in the project process. The most important thing in project management is to balance the interests of all parties and to eliminate, as far as possible, the negative impact of the project on the project by the parties involved.

287 Accomplishment of the twelve principles of the Accompaniment Declaration (1-4)

1. Our highest objective is to meet client needs through the early and sustained delivery of valuable software; and, even at an advanced stage of project development, requests for changes are welcome. The agile process changes by hugging. (c) In the course of the project, business and developers work together on a daily basis;

288 Agile Manifesto Principles 5-8

5. Build projects around motivated individuals; give them the environment and support they need, and trust them to get the job done. 6. The most effective method of communication within and across teams is face-to-face conversation. 7. Working software is the primary measure of progress. 8. Agile processes promote sustainable development, with sponsors, developers, and users able to maintain a steady pace.

289 Accomplishment of the Twelfth Principle (9-12)

Continued attention to technological excellence and design can help to increase agility; This is an art; 11, the best structure, demand and design come from the team; 12, the team has to review and reflect on how it can be more effective and adjust its behaviour accordingly.

290 SCRUM's three-three-four

Three roles: Product Online, Product Manager; Scrum Master, Team Leader; Self-organized Team Self-organizing Teams; Three items: Product Boxlog, Product List; Sprint Backlog; Incremental Deliverable; Four meetings: Sprint Planning; Over-the-counter Review; Sprint Review; Over-the-Return Review; Daily Post.

291 Product Owner

The agility consists mainly of three roles: Product Owner, Scrum Master, and Project Team. Product Owner: Mainly responsible for determining the functionality of the product and meeting the required standards, maintaining the list of items of the product, specifying the content of the delivery of the software and having the authority to accept or reject the results of the development team.

292 Scrum Master

The three players in the Scrum Framework (Scrum Master) are primarily responsible for the smooth implementation and conduct of the entire Scrum process in the project and the removal of communication barriers between clients and development efforts, which allow clients to drive development directly. These include effective application functions for service teams, teaching teams, protection teams, and steering Scrum.

293 Scrum Development Team

Mainly responsible for the development of software products under the Scrum process, the number of people is contained in about 3-9 (Product Owner, SMs are not included in the number of people, unless they participate in the execution of the sprint list), and teams are authorized to organize and manage their work. Each member may be responsible for different technical aspects, but each member must have a strong self-management capacity and be responsible for the entire development team. It provides the team with a path to success, failure, adjustment, improvement.

294 Sprint Planning

Purpose: To determine the deliverables of this Sprint and how to achieve the objectives. Characteristics: Marks the beginning of Sprint; Determines which user stories will be included in the current anatomy; and breaks up the task to collect the task for members of the time estimation team; Product Owner must prepare an updated and sorted list of backlog items for the iterative planning meeting; For Sprint for a month, Sprint will normally not exceed eight hours.

295 Agile - Daily Standup

Definition: Daily meetings to communicate the results within the team and to describe any obstacles. (a) Not more than 15 minutes; What am I gonna do today? Are there any obstacles? Two counter-models: turn into status reports; the station will be designed to detect problems, not solve them.

296 Sprint Review

Definition: A meeting of teams and Product Owner to review the Sprint outcome at the end of the Sprint. Controlled within four hours. Purpose: The team demonstrates the functionality performed in the Sprint (using a relatively real environment to the extent possible) to Product Owner and receives their comments, suggestions and evaluations to review the volume of product increments delivered and adjust product backlog items as required. Results of the review: A revised list of backlog items for products, identifying those that are likely to be next.

297 Sprint Retrospective

Definition: A retrospective session on team self-continuing improvements at the end of Sprint. Usually after the Sprint Review, before the next Sprint planning meeting. A month's sprint does not exceed two hours. Purpose: To take stock of the experiences and problems in this iterative period; To identify and sequence key aspects of potential subsequent improvements; To develop an improved workplan.

298 Agile-servant leadership

Servant leadership in agile management: Definition: an approach to team empowerment. The practice of team leadership through team service focuses on understanding and attention to the needs and development of team members and aims to achieve the highest possible performance of the team.

299 Agile - Product Backlog

The backlog is an orderly list of all jobs that is presented to the team as a story. The greater the value, the higher it is. The product owner produces a product road map showing the expected series of deliverables. The product owner re-engineered the road map based on the actual results of the team. The product owner worked with the team in the iterative meetings to prepare stories for the upcoming iterative period and refine enough stories. The group is briefed on the creativity of stories, potential challenges or issues.

300 Three Scrum Artifacts - Sprint Backlog

The Sprint Backlog defines the Sprint Goal and the specific tasks to be completed during the Sprint. It should be visible to the team. Tasks are not assigned top-down; they result from team discussion and individual selection. Remaining work is updated daily. The Sprint Backlog is produced during Sprint Planning and belongs to the team, which may add, remove, or revise tasks. If the team agrees, some items may first be estimated at a high level.